SF LIVE SUITE

REFERENCE MANUAL

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31 PLUGINS FOR PERFORMING HYPNOTIC / DUB TECHNO IN ABLETON LIVE
10 INSTRUMENTS 12 EFFECTS 9 MASTER AND MIX TOOLS

VERSION 0.2 NOVEMBER 2026
STRANGE FREQUENCIES

CONTROL RANGES AND DEFAULTS IN THIS MANUAL ARE GENERATED FROM THE PLUGIN CODE.

CONTENTS

1 HOW THE SUITE WORKS 4

2 SONGS AND PRESETS 7

3 POCKET: KICK-AWARE DUCKING 8

4 ALLOC: THE MIX ASSIGNER 10

5 SOUND AND LEVELS 12

6 SETTING UP LIVE 13

7 THE SHOW TEMPLATE 15

8 INSTRUMENTS 26

SF KICK 26

SF PERC 30

SF RUMBLE 34

SF BASS 37

SF LEAD 42

SF PAD 47

SF VOICE 52

SF STRIKE 55

SF DECK 58

SF CONTROL 61

9 EFFECTS 64

SF STRIP 64

SF FILTER 67

SF GATE 68

SF DENSITY 69

SF DELAY 71

SF REVERB 73

SF TERROIR 75

SF CAPTURE 80
SF DRIVE 82
SF RADIO 83
SF REPEAT 85
SF DUCK 87
SF ENERGY 88

10 MASTER AND MIX 90
SF ISOLATOR 90
SF BUILD 91
SF GLUE 94
SF TONE 95
SF LOWEND 96
SF LIMIT 97
SF ROOM EQ 98
SF ALLOC 100
SF TAP 103

11 APPENDIX 104

1 HOW THE SUITE WORKS

The suite is built for one kind of performance: very simple note content, often one note in a pattern that never changes, with all the variation played on knobs. Everything follows from that. Every plugin is designed to be played, not programmed, and every plugin talks to the others.

THE RIG

One Live set that never changes. The track layout is fixed: one track per role (kick, percussion, rumble, bass, lead, voice, pad, deck). The OXI One MkII sequences everything; the OXI E16 plays the knobs. Songs are not projects: a song is a stored set of values in every plugin, recalled on the bar. You switch songs without loading anything.

THE PERFORMANCE ROW

The first eight controls of every plugin are its performance row: the first eight parameters Live shows for the device. On the E16 a track has a page of its own, with these and more of its instrument: one page plays one track.

THE PLUGIN WINDOW

Every plugin's window is a narrow strip, so several sit side by side. Top to bottom it goes from what you play to what you set once:

— **Header:** SONG, the OUT meter (FAULT, PANIC), PRESET, STORE, LOAD.

— **E16:** the plugin's page of the E16 scenes (PERF or MIX, named beside the page), as the E16 shows it: its own encoders lit where they sit, the others dark. The rings fill as the E16's do (16 LEDs, from just left of the bottom, clockwise) in the scenes' colours: white, light blue for a level, green for SLOW ALL. A press sits under its knob (THROW, FREEZE, DROP, ROLL, GO); click it. A plugin the scenes have on several tracks (LEAD, PERC, ENERGY) finds its encoders by the track's name, which only the VST3 knows; without it, a line says so and its knobs go to PERFORM. A plugin the scenes don't use has no grid.

— **PERFORM:** the rest of the performance row, and what else is played (DECK's PLAY and CUE, CONTROL's PANIC).

— **The display**, then **DETAIL**: the sound's other controls, MODEL first.

— **ENGINE** (BASS, LEAD, PAD): the controls one model has of its own, folded: FM 4-OP's operators and WAVETABLE's TABLE.

— **LAYER** (KICK, BASS, LEAD, PAD): the second layer, LAYER MODEL, LAYER, SPLIT and SPLIT HZ, folded, how it is set on its title line.

— **MOTION**, **SYSTEM** (SLOW, MACRO) and **SETUP** (how the plugin sits in the set: ALLOC, KEY FOLLOW, DRY ...) come folded, their settings on their title line. Click a group's title line to open or fold it; the plugin keeps it so, with the set.

— **Taller than the screen** (some strips pass 1000 pixels with every group open): the window stops at the screen's height and the strip scrolls under the header, by the thin bar at the right or the wheel. While it scrolls, the wheel scrolls and turns no knob; a group you open at the bottom scrolls into view.

— **On every knob:** green, the sound travelling to where you set it (SLOW), and the ringed LED is where it is going; a green dot outside the ring, where MACRO or MOTION have it now; the value in cyan while you touch it. A knob whose middle is neutral (FILTER, TUNE) fills from the top either way, except in the

E16 grid, which fills as the E16 does.

DENSITY, GATE, FILTER

Three controls cover most of what changes in a track, and they appear wherever they make sense. **DENSITY** is how many events happen (ghost notes fill the gaps). **GATE** is how long each event lasts. **FILTER** is what each event sounds like: a bipolar DJ filter, centre open, left low-pass, right high-pass. Play two against each other: density up with gate down is a tight, nervous build; density down with gate open is spacious dub; density up with the filter closed is tension waiting to be released.

SYSTEM CONTROLS ON EVERY PLUGIN

At the bottom of every plugin's window, folded: MOTION (two modulators) and SYSTEM (SLOW, MACRO). They are the same on every plugin and are not repeated in the module tables.

CONTROL	RANGE	WHAT IT DOES
SLOW	CONTROL, 1-64 BAR, OFF, 128 BAR	Slow mode. Every performance-row knob, and the instruments' COLOR, DETUNE, ATTACK and SPACE, travels to where you set it over exactly this many bars, however far you moved it; song recalls glide too. Other knobs follow in about 30 ms; menus and switches (MODEL, CHORD) change at once. CONTROL (the default): SF CONTROL's SLOW ALL sets the length. A move keeps the length it set off with. While travelling, the knob's ring is green and walks with the sound, the ringed LED is where it is going, and its value shows in green.
MACRO SRC	OFF, MACRO 1-6	Which SF CONTROL macro this plugin follows.
MACRO AMT	-100 to +100 %	How strongly and in which direction. Each plugin decides which of its knobs respond (the ones marked M in the tables).
M1 / M2 SOURCE	OFF, SINE, TRIANGLE, SQUARE, RAMP UP, RAMP DOWN, RANDOM, DRIFT, KICK, NOTES or INPUT, STEPS, ENVELOPE	Two modulators (MOTION). Waves, RANDOM (stepped), DRIFT (smooth) and STEPS (eight random steps, looping: a pattern) move around the knob; ramps rise or fall from it and restart every RATE; ENVELOPE jumps up and falls away every RATE, a pump on the grid; KICK follows SF KICK's hits; NOTES (instruments) follows velocity; INPUT (effects) follows the input level.
M1 / M2 RATE	1/16 to 64 BAR, odd	Cycle length, locked to Live's song position; the odd ones run against the bar (MOTION, below). KICK, NOTES, INPUT: the release.
M1 / M2 TARGET	OFF or any knob	The control the modulator moves.
M1 / M2 DEPTH	-100 to +100 %	How far, as a share of the target's whole range. Negative turns it upside down.

MOTION

Two modulators in every plugin, M1 and M2. Pick a SOURCE, a RATE, the TARGET knob and a DEPTH. Everything is locked to Live's song position: every plugin's modulators move together, a 64-bar ramp lands on the 64th bar, and a set replays the same way every time, RANDOM and DRIFT included (their seeds are saved). Modulation adds to the knob, the macro and SLOW; a green dot outside the knob's ring shows where it is. Steps (SQUARE, RANDOM, a ramp restart) are smoothed over 3 ms. MOTION is part of songs and presets.

— **Hypnotic movement:** DRIFT on a filter at 8 or 16 BAR, DEPTH 10-20 %: it never repeats, so neither does the loop.

— **Long builds:** RAMP UP at 32 or 64 BAR on ENERGY or a filter; turn DEPTH to set how far it goes.

— **Polyrhythm:** the odd RATEs (3/16, 5/16, 7/16, 1/4D, 1/2D, 1/4T, 1/2T, 3, 5, 7 BAR) turn against the 4/4: a 5/16 SINE on a filter meets the bar again every 5 bars.

— **Kick-driven:** KICK on a bass's DRIVE or a pad's CUTOFF: every kick pushes it, RATE sets the fall.

— **Played:** NOTES on an instrument (velocity) or INPUT on an effect (level) makes the sound respond to how hard it is played.

KEY

Write parts in C. SF CONTROL's KEY (stored per song) then plays them in the song's key on every SF instrument: BASS, LEAD, PAD and VOICE transpose by the nearest interval (-6 to +5 semitones); KICK, RUMBLE and STRIKE land their own pitch on the key root, so the low end is always in tune with the track. Notes already sounding follow a KEY change. KEY FOLLOW on each instrument opts out. OFF changes nothing, and without SF CONTROL running every instrument plays in C.

Mind the low register. KEY moves parts up to 6 semitones either way, and KICK, RUMBLE and STRIKE snap to the nearest root, so the low end can land high or very low: a kick body at 50 Hz becomes 69 Hz in C# and 37 Hz in D. Check every song's KEY on a sub. TUNE (stored per song) moves the kick from the snapped root: -12 takes the 69 Hz kick an octave down, +7 or -5 puts it on the fifth. A bass too deep: move the part up an octave on the OXI.

NOTHING CAN HURT THE PA

Every knob is smoothed, so 7-bit MIDI never steps audibly. Resonance is level-compensated. Feedback above 100 % saturates instead of exploding. Every output is clamped at +12 dBFS. A plugin that produces an invalid value, or sits at the clamp for 250 ms (a runaway), outputs silence, resets itself, shows **FAULT** in red and carries on. **PANIC** on SF CONTROL fades every SF plugin out over 5 ms and clears every voice, tail and delay line: for a runaway loop, a stuck note or a wrong recall. While held, everything stays silent (no input, no new notes; DECK keeps its place; every header shows PANIC in red); let go and it fades back in over 5 ms. A set never reopens with PANIC held. Grab any knob in the dark.

THE COLOUR CODE

The plugins are black, white and grey. Colour only carries information:

	COLOUR	MEANS
	RED	Clip, FAULT (the plugin reset itself), missing file, true peak at the ceiling.
	AMBER	Conflict or warning: an ALLOC conflict, a deck armed and waiting, RETURN KNOB on BUILD.
	GREEN	Active or locked: bus connected, kick reporting, a song recall or a drop waiting for the bar, a value gliding (SLOW).
	CYAN	The value of the control you are touching.
	E16 RINGS	In the E16 grid at the top of a window, each ring has its colour on the E16, by role: white an instrument's own sound (1-8), red a track's GATE or DECAY, DENSITY and FILTER (9-11) and its NRG, light blue a level or a send, pink NRG's ACID, DIST and WARP, violet NEXT SONG and BUILD, green SLOW ALL, and a knob while it glides.

2 SONGS AND PRESETS

Every SF plugin stores 16 song slots of its own settings, saved with the Live set. SF CONTROL chooses the song; every plugin recalls its slot on the same bar line while Live plays (at once when stopped).

SAVING A SONG

— Set every plugin how the song should start.

— Press **STORE ALL** on SF CONTROL. Every SF plugin stores its current values into the song in force (SONG).

— **Into another song:** choose it in **STORE TO** on SF CONTROL, then press STORE ALL. That song isn't recalled first, so nothing you hear changes; it becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG. Choosing the song on SONG instead would recall it, and what you had set would be gone before you stored it.

— If a song change is waiting for its bar (WAIT BAR) when you STORE TO, the change still lands and stays the song in force.

— STORE in any plugin's header stores just that plugin into the selected song.

RECALLING A SONG

— Send Program Change on MIDI channel 16 to SF CONTROL (program 1 = song 01). Or turn SF CONTROL's SONG.

— SF CONTROL decides the bar line and sends it with the song (and the song's KEY), so every plugin changes on the same bar whatever order Live processes the tracks in. A Program Change up to a sixteenth after a bar line counts for that bar (a sequencer sends it on its downbeat, and one clocked by Live plays a little after Live's), a turn of SONG up to 30 ms after it; later in the bar, for the next one. Inside an Arrangement loop that never reaches that bar, every plugin changes on the next bar line after the loop jumps back.

— The header of every plugin shows SONG nn, and WAIT BAR in green while the recall is waiting for the bar line.

— Recalled values are reported to Live, so the E16 rings jump to the new song's settings. A recalled knob is yours again the moment you turn it.

— A plugin with nothing stored in that song stays as it is.

— With SLOW ALL (SF CONTROL) or a plugin's own SLOW on, the recall glides instead of jumping: a song change becomes an 8- or 16-bar transition.

— Arrangement automation wins over recalled values, as on any device.

— A preset loaded right after a recall (Live's browser or undo, or the PRESET menu) wins, whatever its values. A song change still waiting for its bar, or a STORE not yet made, in that plugin when a preset from Live comes in is dropped there: ask again.

VST3 AND PROGRAM CHANGE

VST3 plugins don't receive MIDI Program Change as MIDI. SF CONTROL therefore lists 16 programs (SONG 01-16); Live 12.2 and later turn a Program Change on the track into a program change for the VST3, older versions don't. The AU version reads Program Change directly from MIDI in any version. If song changes don't arrive in the VST3, use the AU.

PRESETS

— The **PRESET** button in every header opens the menu: factory presets grouped by style (DUB, ACID, HYPNOTIC ...), your own presets, SAVE USER PRESET and SHOW PRESET FOLDER. The button shows the preset that is loaded.

— A preset sets every sound control of the plugin, MOTION included, and Live is told about every change, so the E16 rings follow. It never touches SLOW, MACRO, ALLOC, ROLE, a source setting or KEY FOLLOW.

— 219 factory presets across 27 plugins. Use them as starting points, then store the result into a song.

— **RANDOMISE THE UNLOCKED** (in the PRESET menu) sets every unlocked sound control anywhere in its range, MODEL and other menus included; **NUDGE THE UNLOCKED** moves the unlocked knobs a little (up to 15 %) from where they are. Right-click a knob to lock or unlock it (a small padlock by its ring). At first LEVEL, FILTER, DENSITY, MIX, TUNE, SHIFT, ACID, WARP and ENERGY are locked, so a random patch stays in key and at its level. The locks are saved with the set.

— User presets are files in Music/SF Live Suite/Presets/<PLUGIN>. Copy the folder to take them to another computer.

3 POCKET: KICK-AWARE DUCKING

SF KICK reports every hit to the other SF plugins: when it hit, how hard, its pitch sweep, how long it rings and how its filter is set. Any module with Pocket carves a kick-shaped hole in itself. No sidechain routing.

TWO KNOBS

DEPTH (or PUMP, DUCK) sets how deep. **SHAPE** is one knob from transparent to pumping:

— **Left:** a narrow cut that follows the kick's pitch sweep down to its fundamental. The bass keeps its body; you feel more space without hearing a duck.

— **Middle:** the low band (below about three times the kick's fundamental) ducks for the kick's length.

— **Right:** full-band pump, release tied to the tempo.

IT FOLLOWS THE KICK

— **Velocity:** ghost kicks get shallow pockets.

— **DECAY:** lengthen the kick and every pocket lengthens.

— **Pitch:** the cut follows the sweep wherever TUNE and PUNCH put it.

— **Kick FILTER:** high-pass the kick and the low pockets close: the bass and rumble take over the sub. Bring the kick's low end back on the drop and the pockets open on the downbeat.

— **Rolls:** on transparent settings fast hits merge into one dip; on pumping settings each hit pumps.

— **Safety:** if kick tails overlap so much that the pocket would never let go, it shrinks to the attack.

— **Breakdowns:** when the kick stops, pockets close; when it returns they fade back in over a beat.

WHO HAS POCKET

BASS, LEAD, PAD, VOICE (POCKET DEPTH / SHAPE), RUMBLE (PUMP), DELAY and REVERB (DUCK, on the wet signal only), DUCK (on anything), TAP (on any track). SF DUCK can also follow a sidechain input instead of the kick.

TIMING

Pocket looks 512 samples ahead (about 11 ms at 48 kHz) so a kick report is always there before it's needed. Live compensates the delay. Keep Live's buffer size at 512 or less. If the pump lands early or late in your setup, trim POCKET OFFSET on SF CONTROL (+/-20 ms) once, by ear.

AFTER A DROP

SF BUILD's DROP DUCK (off by default, up to 12 dB) makes every Pocket stage dip under each kick for DUCK TIME after the drop (default 1 bar), whatever its DEPTH, so the first kicks of the drop land on top of the parts and the tail coming back. It follows the kick: back within half a beat of each hit, and easing out over the second half of DUCK TIME. It starts with the first kick on the drop's bar line: a roll into the drop doesn't duck. TAP and STRIP set to ROLE KICK, PERC or DECK don't dip, like SF KICK, PERC, STRIKE and DECK.

LOOPS, RESTARTS AND JUMPS

Everything follows Live's song position. When it jumps (arrangement loop brace, stop and restart, clicking to a new position), kick reports from before the jump are ignored, so nothing ducks for a kick that is no longer playing. RUMBLE re-arms. Playing DECK tracks keep running and are re-anchored by whole beats, so they stay on the beat.

4 ALLOC: THE MIX ASSIGNER

ALLOC keeps sources from piling up in the same frequencies. You decide the order of importance; each frequency band has a priority curve over that order; the display shows what every source keeps and loses. Nothing is decided behind your back.

THE ORDER

— Every SF instrument, STRIP, TAP and DECK reports its energy in ten fixed bands and appears in the SOURCES list (by its Live track name with VST3, otherwise by role: PERC, PERC 2 ...).

— Drag the list into order, top = most important: the order for ALL BANDS. It is saved with the set and only changes when you drag it. New sources are added at the bottom.

— Low bands and high bands usually want different orders (kick and bass matter at 60 Hz, hats and voice at 3 kHz). Click a band to show its order in the list and drag: the band now has its OWN order. LINK TO ALL returns it to the global order.

— Each source applies its own cuts: audio never leaves its track.

THE CURVE (PER BAND)

— **PRIORITY** moves the peak of the curve up and down the list. The source at the peak is protected.

— **DEPTH** sets how hard everything away from the peak is cut, up to MAX CUT. At 0 the band is untouched.

— **Q** sets the width. Narrow: one source owns the band. Wide: the sources next to the peak share it.

— Each knob always does the same thing to the same band, so once set it stays predictable.

DYNAMIC OR STATIC

DYNAMIC (default): a source is cut only while a more favoured source is actually playing in that band. When the bass stops, the pad gets 180 Hz back within a fraction of a second. **STATIC**: the cuts stay on all the time, like a fixed EQ.

READING THE DISPLAY

— Each band is a column; each source a lane, in that band's order and named in the lane. A selected band has a white frame; OWN marks a band with its own order.

— **Grey**: what the source plays in that band after ALLOC. **Amber**: what ALLOC takes away.

— The white line is the curve (right = protected); it turns cyan while you turn one of that band's knobs.

— The name at the bottom of the band is the source the peak sits on.

THE BANDS

BAND	RANGE
1	20-60 Hz
2	60-100 Hz
3	100-150 Hz
4	150-220 Hz
5	220-320 Hz
6	320-480 Hz
7	480-750 Hz
8	750-1500 Hz
9	1.5-4 kHz
10	4-20 kHz

5 SOUND AND LEVELS

The sets are the releases, so every plugin is built to release quality: nothing you hear should come from the implementation. These figures are measured by the offline tests on every build.

GAIN STAGING

Every model of every instrument is calibrated to its role's loudness at default settings, so switching models never jumps the level and a rig with every LEVEL at its default is already a balanced rough mix with headroom into GLUE and LIMIT. SAMPLE models play the file at its own level.

ROLE	TARGET (LUFS, INTEGRATED)	ROLE	TARGET
KICK	-15	LEAD	-20
PERC	-22	PAD	-22
RUMBLE	-20	VOICE	-20
BASS	-18	STRIKE	-20

NO ALIASING

Every stage that distorts runs oversampled, all the time (switching an oversampler in and out would click), and every drive knob fades in from exactly clean: synth voices at 2x (the ladder, MS20-TYPE and SYNC models at 4x), KICK and PERC at 2x, DRIVE and STRIP at 4x, the drive in FILTER, ENERGY and VOICE, the saturation inside the DELAY loop, RUMBLE's GRIT, LOWEND's SUB CLIP, STRIKE's fracture and TONE at 2x, LIMIT's clipper at 4x. The filters reject images and aliases by 112 dB or more and are flat to 20 kHz. Oscillators are band-limited (wavetables per octave, corrected hard sync): measured alias products sit 77 to 105 dB below the note.

THE MASTER

— SF LIMIT measures true peak (the level between samples that a DAC or a codec will produce) at 8x, and its output never exceeds CEILING. Use -1 dB for venues and releases.

— Its meter follows ITU-R BS.1770-4 / EBU R128 exactly: MOMENTARY (400 ms), SHORT-TERM (3 s), INTEGRATED (gated, over the whole set since RESET). It passes the EBU Tech 3341 level, gating and short-term cases within 0.02 LU.

— SF LIMIT AUTO keeps the loudest moments of the last minute at TARGET and holds through breakdowns, so the drop comes back where it was. AUTO MODE DOWN ONLY (the default) never lifts the set above unity.

— SF TONE voices the whole set (LOW, MUD, PRESENCE, AIR, TILT); SF ROOM EQ corrects the room, per venue. Keep them separate: TONE is the record, ROOM EQ is the venue, and only the venue hears ROOM EQ (below).

ANY BUFFER SIZE, ANY SAMPLE RATE

Every plugin processes in 64-sample steps on a fixed grid of the audio stream, whatever Live's buffer size, so a set sounds exactly the same at buffer 64 and buffer 2048 (bit for bit) and at 44.1, 48 or 96 kHz. Song recalls land on the exact sample of the bar line.

RECORDING A SET

— Record the Main output (after SF LIMIT, before SF ROOM EQ) at the session's sample rate, 24-bit or 32-bit float: an audio track REC with Audio From = Main, Post Mixer, Monitor Off, armed. Live records what comes into a track, before its devices.

— The room's correction stays out of the release: SF ROOM EQ sits on the PA track (see Setting up Live), or do the venue EQ in the audio interface's own mixer, after Live.

— Press RESET on SF LIMIT before you start: INTEGRATED then reads the loudness of the whole set.

— MOTION's RANDOM and DRIFT are seeded and the seeds are saved with the set, so they move the same way every time the set is played.

6 SETTING UP LIVE

TRACKS

WHERE	WHAT
MIDI track	SF CONTROL, input from the OXI One. Makes no sound.
MIDI tracks	SF KICK, SF PERC (2-3), SF RUMBLE (no clips needed), SF BASS, SF LEAD (1-2), SF VOICE, SF PAD, SF STRIKE, SF DECK.
Groups	SF ENERGY on each group, PROFILE to match (DRUMS, BASS, MUSIC, VOICE).
Returns	SF DELAY, SF REVERB, SF DENSITY, each with DRY off (the return gives only the effect).
Main (master)	SF ISOLATOR > SF BUILD > SF GLUE > SF TONE > SF LOWEND > SF LIMIT. SF ALLOC anywhere on it. Its output to an output pair you don't use.
Audio track PA	Audio From = Main, Post Mixer (the Main output), Monitor In, SF ROOM EQ, then SF LIMIT as a safety (CEILING -0.5 dB, CLIP 0 %, AUTO off), Audio To = the PA outputs, fader at 0 dB. The venue hears ROOM EQ.
Audio track REC	Audio From = Main, Post Mixer, Monitor Off, armed: records the set without ROOM EQ.
Audio track ROOM	Audio From = the room mics (a stereo pair), Monitor In, SF CAPTURE, Audio To = the outputs a speaker can go on (the headphones), sends disabled. SF TERROIR captures the place from here (SF CAPTURE); armed, it records the room with the set.
Live instruments / hardware	SF STRIP last on the track: the suite's performance row, Pocket and ALLOC for any sound.
Stems	SF TAP with ROLE set (no sound of its own), or SF STRIP to play them.

THE E16

— Map with Live's MIDI Map (Cmd-M): unfold the plugin's parameter panel (the triangle in its title bar), click the parameter there and turn the E16 knob. A parameter that isn't listed: Configure, touch it in the plugin window, Configure off. Turn to a PERF page before you click a parameter: a PERF page sends TRACK FOCUS's CC as it comes up, and Cmd-M would map the selected parameter to it.

— In Live's MIDI preferences turn **Remote** on for the E16's Port 1, input and output (Ports 2 and 3 off). Live then sends values back, so the rings follow song recalls and ALLOC slot changes and the knobs hold the new values.

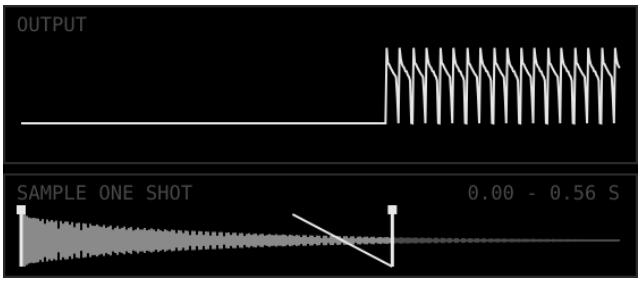
— Absolute 0-127 CC is right; the plugins smooth every step. Chapter 7 has two ready-made E16 scenes for the show template, PERF and MIX, and the show set with every address already mapped.

SAMPLES

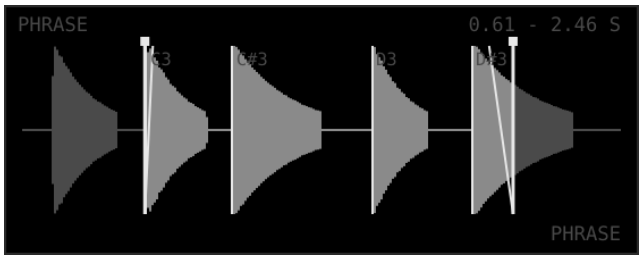
Every instrument has a SAMPLE model. Drop an audio file on the plugin window or choose LOAD > FILE.... The sample is analysed (root note, decay, onsets) and embedded in the Live set, so the set travels without the file.

RESAMPLE (LOAD > RESAMPLE 1, 2 or 4 BARS): from the next bar line, what the plugin plays becomes its sample, and a WAV in Music/SF Live Suite/Resamples. LOAD reads WAIT until the bar, then REC. Hold a chord on PAD and resample it, then play it on SAMPLE, GRANULAR, or WAVETABLE with TABLE on SAMPLE (eight frames from the sample, COLOR moves across them).

The part of the sample that plays is set like an Ableton clip's: START and END, with FADE IN after START and FADE OUT before END (the SAMPLE group, folded at first). With a sample loaded the display shows its waveform, the part that plays bright, the fades as ramps and START and END as lines: drag either line (it takes the nearer one; near the sample's ends it snaps to them). VOICE draws them on its phrase. The instrument's own shaping (DECAY, ATTACK, GATE) still works on each note on top. The fades run in the sample's own time: played an octave up, they pass in half the time. A cut inside the waveform always fades for at least 1 ms, so it never clicks. At their defaults (0 %, 100 %, 0 ms, 0 ms) the sample plays whole, exactly as before these existed. Presets leave them alone (a preset has no sample); songs keep them like any knob.



SF KICK with a one-shot: END at 62 %, FADE OUT 150 ms. The part that plays is bright.



SF VOICE: START on the second word, END in the last. The slices (C3, C#3 ...) are cut from the part between.

FIRST SESSION CHECKS

— SF CONTROL shows **BUS SHARED**. Play SF KICK: it shows **KICK REPORTING**.

— SF KICK + SF BASS with POCKET DEPTH up and SHAPE right: the pump lands exactly on the kick.

— SF ALLOC on the master: every SF source appears in the list; raise DEPTH on a band where two of them overlap and watch the amber.

— Store two songs; switch with Program Change: the E16 rings follow. With SF FILTER on tracks above and below SF CONTROL, every track changes on the same bar.

— SF DECK: load a track, PLAY A: it starts on the bar, locked to Live's tempo.

— Set KEY on SF CONTROL: BASS, LEAD and PAD transpose, KICK and RUMBLE retune.

— Record a minute on REC with a deep cut on ROOM EQ: the PA has the cut, the recording doesn't.

Chapter 7 is a complete show set built on this layout.

7 THE SHOW TEMPLATE

One Live set for every show, built on fixed roles. The OXI ONE plays the notes and patterns; the E16 plays the sound, the balance and the transitions. One number per track: a track's number is its MIDI channel on the ONE and on the E16, and its page in the E16 scene PERF, so Live, the ONE and the E16 read the same way.

The set is built: **controllers/live/SF SHOW.als** in the repository holds every track, group and return below, every SF plugin with its starting settings, the routing, and every address of both E16 scenes mapped. Every instrument's MIDI From is set to its OXI One port and channel; PA and REC take the Main. Set Live's own preferences once per computer (its README, section 1: the MIDI ports, audio, Takeover Mode), check its outputs against your interface before any sound reaches a PA, play, and save it as your template.

TRACKS

TRACK	DEVICES	MIDI FROM
CONTROL	SF CONTROL	ONE port A, ch 16
1 KICK	SF KICK; SF ENERGY	ONE port A, ch 1
DRUMS (group)	SF ENERGY, PROFILE DRUMS; SF REPEAT	
2 CLOSED HAT	SF PERC: closed hat; SF ENERGY	port A, ch 2
3 OPEN HAT	SF PERC: open hat or ride; SF ENERGY	port A, ch 3
4 CLAP	SF PERC: clap or rim; SF ENERGY	port A, ch 4
5 PERC	SF STRIKE; SF ENERGY	port A, ch 5
LOW (group)	SF ENERGY, PROFILE BASS	
6 BASS	SF BASS; SF ENERGY	port A, ch 6
7 RUMBLE	SF RUMBLE, SOURCE KICK BUS; SF ENERGY	none: it follows the kick
MUSIC (group)	SF ENERGY, PROFILE MUSIC	
8 LEAD	SF LEAD: ACID 303-TYPE or another mono lead; SF ENERGY	port A, ch 8
9 CHORD	SF LEAD: chords and stabs; SF ENERGY	port A, ch 9
10 PAD	SF PAD, often GRANULAR; SF ENERGY	port A, ch 10
11 VOICE	SF VOICE: phrases, radio fragments, choir; SF ENERGY	port A, ch 11
BRIDGE	Live audio clips	
DECK	SF DECK	port B, ch 7 (hot cues; optional)
LIVE BUS	SF ISOLATOR > SF BUILD > SF GLUE > SF TONE > SF LOWEND; SF ALLOC (off at first)	
PA	SF ROOM EQ > SF LIMIT (safety)	audio: Main, Post Mixer
REC	nothing	audio: Main, Post Mixer
ROOM	SF CAPTURE	audio: Ext. In 1/2, the room mics
A DUB (return)	SF DELAY, DRY off	
B SPACE (return)	SF REVERB, DRY off	
Main	SF LIMIT	

— Tracks are named by their role, not their instrument: a track keeps its name, number, channel and E16 page when another instrument goes on it. Map its page to the new instrument (Cmd-M) and nothing else moves.

— Every instrument track ends in SF ENERGY, with the profile of its part: its ENERGY knob on the E16's NRG page. The groups have one each too.

— The kick stays outside DRUMS and LOW, so the drum group's REPEAT and ENERGY never take it, and the bass and the rumble fade together while the kick holds the floor. In each section decide whether BASS or RUMBLE carries the sustained low end: one of two knobs then steers it.

— The PERCs are separate instruments: CHOKE works inside one track only, so a closed hat doesn't cut the open hat on its own track. Write the open hat's rhythm and DECAY for that.

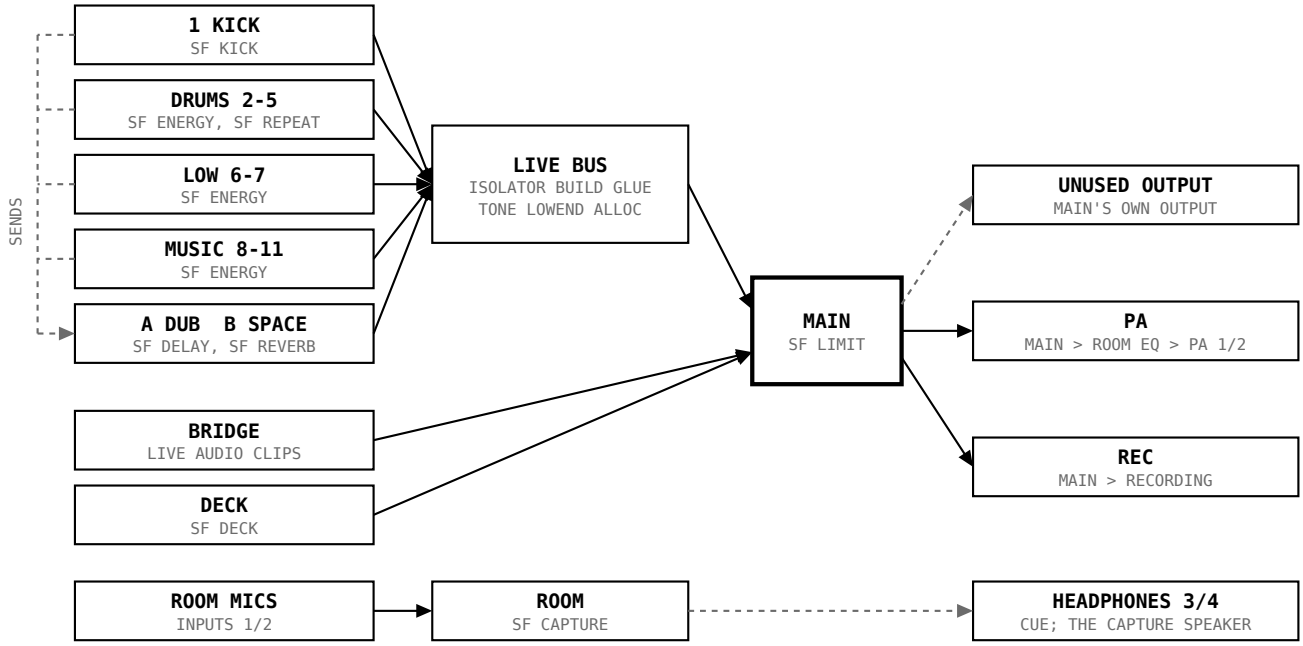
— CHORD: if the ONE sends whole chords, set the plugin's CHORD to NONE; if it sends single notes, CHORD (or the model's own chord) voices them. Not both.

— LEAD's ACID 303-TYPE slides when notes overlap, always over 60 ms like a 303: write legato notes for slides. GLIDE doesn't change it.

— VOICE with a phrase loaded: C3 plays slice 1, C#3 slice 2 and so on; below C3, the whole phrase. Sequence it as phrase choreography, not as a pitched sampler.

— Keep the names: SF ALLOC lists its sources by Live track name, and SF LEAD, PERC and ENERGY find their E16 knobs by it (VST3).

ROUTING



TRACK	INPUT / MONITOR	OUTPUT
Instrument tracks	MIDI From as above, Monitor In	their group; 1 KICK to LIVE BUS
Groups, returns A and B		LIVE BUS
LIVE BUS	the tracks routed to it, Monitor In, every send disabled	Main
BRIDGE	clips, Monitor Off (clips still play)	Main
DECK	MIDI From port B ch 7, Monitor In	Main
Main		an output pair that goes nowhere
PA	Audio From Main, Post Mixer, Monitor In, sends disabled	the PA outputs
REC	Audio From Main, Post Mixer, Monitor Off, sends disabled	Sends Only
ROOM	Ext. In 1/2 (the room mics, a stereo pair), Monitor In, sends disabled	the headphone outputs: a speaker on them for a capture
Cue	Live's Cue Out; DECK's CUE track	the headphone outputs

— Main is the set. Exporting or resampling it gives the release mix; the venue's correction (ROOM EQ) exists only on PA, so the recording never hears it.

— **Disable every send on LIVE BUS** (right-click a send, Disable Send). The returns feed LIVE BUS, and Live can't compensate a return's delay on a track whose matching send is active, even at zero

(Ableton's Delay Compensation FAQ).

— Never send LIVE BUS to PA: BRIDGE, DECK and SF LIMIT would be left out of the PA, and REC would record only BRIDGE and DECK.

— Pull LIVE BUS down and the live instruments and their effects go while BRIDGE and DECK keep playing. The audio interface's output knob is the PA's volume and the emergency stop: turn it down and the room goes quiet while REC keeps recording. Nothing that would stop the set is in Live or on the E16.

— BRIDGE is for transitions and regrouping. It still needs Live: it is not a backup for a computer failure.

— PA's SF LIMIT is a safety: CEILING -0.5 dB, no clip, AUTO off. The set's peaks are already at -1 dBTP, so it only catches what the room EQ's phase pushes past. Keep PA's fader at 0 dB: the PA's level is the interface's.

— ROOM is the room: armed, it records the mics with the set, for the mix. Its SF CAPTURE never passes them, so the set never hears the mics; for SF TERROIR it plays the sweep to a speaker on the headphone out (transport stopped) and records the place (SF CAPTURE).

THE MASTER DEVICES

— Start GLUE, TONE and LOWEND gently: the set doesn't need every processor working all the time. Soundcheck the busiest passage of the night and set LIMIT's INPUT for 1-3 dB of limiting there.

— LIMIT AUTO off while you set the gain staging, then on, AUTO MODE DOWN ONLY, for shows: it stops the set creeping above TARGET and holds through breakdowns.

— SF ALLOC off at first. Turn it on band by band once the arrangement is known. It measures each source before its LEVEL and before Live's fader, so a source you've faded out still claims its bands: take a source out with its lane mute or its FILTER, or switch its ALLOC off.

THE OXI ONE

Every part plays on port A, on its Live track's number: channel = track. SF SHOW.als listens to each on its own track (Live lists port A as OXI ONE MKII (Port 1), port B as (Port 2)), never All Ins / All Channels.

SEQUENCER	MODE	OUTPUT	TRACK
1 DRUMS	Multitrack: a track per drum, the kick on the bottom row	1A-5A	1 KICK, 2 CLOSED HAT, 3 OPEN HAT, 4 CLAP, 5 PERC
2	Mono	6A	6 BASS
3	Mono	8A	8 LEAD
4	Mono to start; Chord or Poly for whole chords	9A	9 CHORD
5	Mono to start; Poly for chords	10A	10 PAD
6	Mono	11A	11 VOICE
7	spare (SF START: Mono)	7B	DECK's hot cues
8	spare (SF START: Mono)	16A	CONTROL (a Program Change, for an AU of it)

— The multitrack's rows, bottom up: 8 KICK 1A, 7 PERC 5A, 6 CLAP 4A, 5 OPEN HAT 3A, 4 CLOSED HAT 2A. 7 RUMBLE needs no lane: SOURCE KICK BUS follows SF KICK.

— KICK lane on C1: with KEY TRACK on, any other note transposes the kick (C1 = TUNE), so C1 is right either way. PERC plays any note.

— KEY OFF on SF CONTROL to start with: the ONE's notes set the pitch, and KICK and RUMBLE are tuned per song with TUNE. Turn KEY on for songs that should change key without reprogramming the ONE

(mind the low register, chapter 1).

— One ONE project per track, with a pattern per section: SPARSE, MAIN, VARIATION, EXIT, each a copy of the startup project SF START (controllers/oxi-one: every sequencer as above, four scenes of simple patterns). The song comes from the E16 (NEXT SONG and GO), never from the project: any song plays with any project's patterns. Program Change stays off on every sequencer.

— When patterns change is set on the ONE (its launch and arranger settings): Live's launch quantization doesn't reach it.

CLOCK, MIDI AND LATENCY

LIVE MIDI PORT	TRACK	SYNC	REMOTE
OXI ONE MKII (Port 1) in: the ONE's port A	on	off	off
OXI ONE MKII (Port 2) in: its port B	on	off	off
OXI ONE MKII (Port 1) out: the clock	off	on	off
OXI E16 (Port 1) in	off	off	on
OXI E16 (Port 1) out	off	off	on
Every other port, E16 Ports 2 and 3 included	off	off	off

— Live is the clock master. On the ONE: sync USB (stopped, Shift + Stop until the header shows USB, not INT); in its config, Start Stop Msgs Ignore No, MIDI Transport Msgs Send Off, MIDI Thru off, Auto Sync on (it starts with Live). Never send clock both ways.

— Live sends its clock only while it plays: start and stop from Live. Started with its own Play, the ONE runs at its own tempo and ignores Live's.

— Audio: 48 kHz, buffer 256 (Pocket needs 512 or less), Delay Compensation on, **Reduced Latency When Monitoring off** (it skips compensation on monitored tracks, so returns and Pocket drift), launch quantization 1 bar, record quantization off.

— The ONE plays a little after Live's beat (the clock goes out, the notes come back: on Nov 7, 40 ms), and nothing needs it on the beat: SF BUILD lands its drop on the kick it hears (LAND ON KICK, with a KICK readout), and Pocket and the effects that cut on the beat (GATE, STRIP, ENERGY's chop, DUCK's GRID, REPEAT, DECK) follow where SF KICK's kicks land. Leave the ONE port's MIDI Clock Sync Delay at 0 (on Nov 7, -70 ms moved nothing). Leave POCKET OFFSET at 0 unless the pump sounds off.

— Every Pocket stage in a chain adds its 512-sample lookahead to the total latency (compensated, but it adds up): no SF TAP or SF DUCK on tracks that already hold SF instruments.

THE E16

Two E16 scenes run the whole rig, in the repository under controllers/e16 with their address map (sf-show-map.csv) and loading steps (README); SF SHOW.als has every address mapped. **PERF** is the show: pages 1-11 are the tracks (page n is track n, titled with its code), page 12 is NRG. **MIX** holds the levels, the returns and the macros. On the E16 every track goes by a three-letter code: KCK, CHH, OHH, CLP, PRC, BAS, RMB, LED, CRD, PAD, VOX, and the groups DRM, LOW, MUS (an ENERGY knob reads KCK NRG, a volume KCK VOL). The E16 is a 4 × 4 grid, numbered row by row from the top left. A track's page:

1 SOUND PRESS: SLOW ALL	2 SOUND ITS OWN	3 SOUND ITS OWN	4 SOUND ITS OWN	ITS SOUND ROWS 1 AND 2
5 SOUND ITS OWN	6 SOUND ITS OWN	7 SOUND ITS OWN	8 SOUND ITS OWN	
9 GATE OR DECAY	10 DENSITY	11 FILTER	12 LEVEL	EVERY TRACK ROWS 3 AND 4
13 SEND A INTO A DUB	14 SEND B INTO B SPACE	15 NEXT SONG PRESS: GO	16 BUILD PRESS: DROP	

LIGHT BLUE RING: A LEVEL OR A SEND. 15 AND 16 ARE THE SAME KNOBS ON EVERY PERF PAGE, NRG INCLUDED.

— **Press 1, on every PERF page:** SLOW ALL (swap): a quick press flips knob 1 to SLOW ALL on SF CONTROL (green ring): how long your next moves take, OFF to 128 BAR; another press flips it back.

— **Press 15, on every PERF page:** GO (hold): calls the song NEXT SONG picked, on the next bar line (nothing if it is already the song in force).

— **Press 16, on every PERF page:** DROP (hold): first press: everything stops on the next sixteenth, the tails ring; second press: the drop on the next kick.

— A knob is easily pressed while it is turned, so every press on the E16 is one the set plays through if it happens by accident. Nothing that would stop the set is on the E16: the PA's emergency stop is the audio interface's output knob.

NRG, page 12: every ENERGY knob, in the tracks' order, then ACID, DIST and WARP.

1 KCK NRG PRESS: SLOW ALL	2 CHH NRG CLOSED HAT	3 OHH NRG OPEN HAT	4 CLP NRG CLAP
5 PRC NRG PERC	6 BAS NRG BASS	7 RMB NRG RUMBLE	8 LED NRG LEAD
9 CRD NRG CHORD	10 PAD NRG PAD	11 VOX NRG VOICE	12 ACID ACID (8 LEAD)
13 DIST DIST (LIVE BUS)	14 WARP WARP (10 PAD)	15 NEXT SNG PRESS: GO	16 BUILD PRESS: DROP

KNOB N IS TRACK N'S SF ENERGY: A PRESS SETS IT BACK TO THE CENTRE (KNOB 1: SLOW ALL). 12-14 ACID, DIST, WARP.

— The ENERGY knobs are a DJ filter per track, plus drive, space, echo and chop: centre is neutral, left takes energy out, right puts it in. Each track's own ENERGY shapes that part. A drum track's ENERGY (PROFILE DRUMS) at zero takes the part out entirely: KCK NRG down, no kick. The groups keep their SF ENERGY, at the centre (MACRO 1 can move it).

- NRG 12-14 [Nov 10]: **ACID** (8 LEAD's SF LEAD: RES, ENV AMT, ACCENT, DRIVE and CUTOFF up together, the 303 squelch), **DIST** (SF DRIVE's MIX on the LIVE BUS: the whole mix from clean to driven, tube into fuzz, its lows under 120 Hz clean and its level held), **WARP** (10 PAD's SF PAD: tape wow and flutter on every voice its own way, more spread and ensemble; on GRANULAR the grains scatter). Each off at zero; their presses do nothing.
- Each instrument's LEVEL belongs to its song; the volumes on MIX's LEVELS page are your balance on the night. Set every LEVEL at soundcheck so a volume at 0 dB gives the intended mix: every volume and send is mapped with a maximum of 0 dB.
- Rings by role, in the OXI App's colour numbers (layout.py holds them): white (34) an instrument's own sound, knobs 1-8; red (63) a track's GATE or DECAY, DENSITY and FILTER (9-11) and its NRG knob; light blue (18) a level: LEVEL, the sends, every volume; pink (50) NRG's ACID, DIST and WARP; violet (9) NEXT SONG and BUILD; green (90) knob 1 on SLOW ALL. Each plugin's window draws its E16 knobs in the same colours. A knob that is travelling (SLOW) turns green and walks with the sound until it arrives (GLIDE RINGS: SF CONTROL's E16 RINGS and the scenes' script; controllers/e16 README).
- **TRACK FOCUS:** turn to a PERF track page and Live selects that track (page 6, BAS: 6 BASS), with the control surface SF E16 installed (controllers/live/SF_E16, into the User Library's Remote Scripts; Settings > Tempo & MIDI: SF E16, Input E16 Port 1, Output None). NRG and MIX's pages leave the selection where it is. With Auto-Hide Plug-In Windows on (Settings > Plug-Ins), the track's open plugin windows come up with its page. The page sends CC 119 on channel 16: keep it unmapped (check_set.py fails a set that maps it).

THE E16: PAGES

PAGE	KNOBS 1-8	KNOBS 9-16	PRESSES
PERF 1: KCK	TUNE / PUNCH / CLICK / DRIVE / PITCH T / LAYER / ECHO SPC / ECHO SWG	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 2: CHH	TUNE / TONE / ACCENT / DRIVE / ECHO SPC / ECHO SWG / ECHO PRB / SPREAD	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 3: OHH	TUNE / TONE / ACCENT / DRIVE / ECHO SPC / ECHO SWG / ECHO PRB / SPREAD	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 4: CLP	TUNE / TONE / ACCENT / DRIVE / ECHO SPC / ECHO SWG / ECHO PRB / SPREAD	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 5: PRC	POWER / DISTANCE / SIZE / MATERIAL / POSITION / TENSION / SPREAD / TUNE	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 6: BAS	CUTOFF / RES / ENV AMT / DECAY / DRIVE / ACCENT / GLIDE / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 7: RMB	SIZE / DARK / GRIT / PUMP / TUNE / PREDELAY / WIDTH / PCKT SHP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 8: LED	CUTOFF / RES / ENV AMT / DECAY / DRIVE / ACCENT / GLIDE / COLOR	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 9: CRD	CUTOFF / RES / ENV AMT / DECAY / ATTACK / SPACE / DETUNE / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 10: PAD	CUTOFF / RES / COLOR / DETUNE / ATTACK / DECAY / SPACE / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 11: VOX	SIGNAL / TONE / TUNE / DRIVE / ECHO / ECHO TME / ECHO FBK / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD	
PERF 12: NRG	KCK NRG / CHH NRG / OHH NRG / CLP NRG / PRC NRG / BAS NRG / RMB NRG / LED NRG	CRD NRG / PAD NRG / VOX NRG / ACID / DIST / WARP / NEXT SNG / BUILD	2-14 centre
MIX 1: LEVELS	KCK VOL / CHH VOL / OHH VOL / CLP VOL / PRC VOL / BAS VOL / RMB VOL / LED VOL	CRD VOL / PAD VOL / VOX VOL / DRM VOL / LOW VOL / MUS VOL / DUB RET / SPC RET	
MIX 2: SPACE	DUB TIME / DUB FDBK / DUB TONE / DUB DRV / SPC SIZE / SPC DCY / SPC DARK / SPC DUCK	-	1 THROW (hold), 2 FREEZE (toggle), 5 THROW (hold), 6 FREEZE (toggle)
MIX 3: MACROS	MACRO 1 / MACRO 2 / MACRO 3 / MACRO 4 / MACRO 5 / MACRO 6 / SLOW ALL / -	RPT LEN / RPT PTCH	9 ROLL (hold)

— On every track page row 3 is GATE (DECAY where the instrument has none), DENSITY, FILTER and LEVEL, and row 4 the track's two sends, NEXT SONG and BUILD: the hand learns one layout. Rows 1 and 2 are the instrument's own, with a swap where a model's character lives elsewhere (GRANULAR: COLOR is grain size, DETUNE is scatter; STAB DUB: SPACE is the ensemble).

— The page titles are the tracks' codes: KCK is 1 KICK's page, CRD 9 CHORD's.

— MIX's LEVELS page has NRG's places: knob n is track n's volume, 12-14 the groups', 15 and 16 the returns'. On SPACE the presses sit under their knobs: 1-4 are DUB, 5-8 SPACE. On MACROS, hold ROLL on 9 and turn LENGTH under the same finger.

— The closed hat's page is the fill page: DENSITY with ECHO SPACE 1/16 turns 1/8 hats into 1/16s, the send-delay trick built in.

THE E16: ADDRESSES AND SETTINGS

— **Channel = track:** 1-11 the instrument tracks (1 KICK ... 11 VOICE), 12 A DUB, 13 B SPACE, 14 the groups (DRUMS, LOW, MUSIC and DRUMS' REPEAT), 15 LIVE BUS, 16 CONTROL. Nothing else uses a track's channel, so a track's page can be mapped to any instrument on that track without touching another.

— **CC = the knob's place + 11 in PERF** (a track's page: CC 12-23), **+ 43 in MIX**; a press its place + 27 in PERF, + 59 in MIX. A track's ENERGY is CC 28 (the groups on channel 14: 28, 29, 30).

— **Mixer:** CC 102 volume, 103 send A, 104 send B, on the track's channel; channel 14: 102-104 the DRUMS, LOW and MUSIC volumes. A parameter on two pages (NEXT SONG, BUILD, SLOW ALL) sends the same address from both: two ways to one mapping. CC 6, 38 and 96-101 (data entry and RPN/NRPN) are never used. The appendix lists every address.

— **Presses:** a hold is a CC, 127 on press and 0 on release, so the parameter is on while the knob is down. A toggle is a note: Live flips a switch on every Note On, so the E16 can never think FREEZE is on while Live has it off. A centre press is the knob's own CC with the value 64. The presses are in the scene files.

— **On the E16:** Auto Transmit off, for both scenes and globally (stored values sent on a page change would pull Live's faders to them), MIDI Thru off. Changing pages or scenes sends no knob's value; a PERF track page sends its track's number (TRACK FOCUS, below).

— **In Live:** E16 Port 1 Remote on, input and output; Ports 2 and 3 off (Port 1 is the one the E16 sends on, and a second Remote port could take a message twice: a toggle would flip twice). Takeover Mode: Value Scaling, so a knob that disagrees with Live after a recall moves the sound smoothly instead of jumping.

— **Mapped once, kept:** MIDI mappings belong to the Live set. SF SHOW.als comes with every address mapped, every volume and send at Max 0 dB and DUB FEEDBACK at 0-70 %; save it as a template (File > Save Live Set As Template) and new sets start mapped. In a set of your own, map once from the appendix and do the same.

— **Map the E16 straight to the SF parameters, not through Instrument Rack macros.** A song recall changes the plugin's value underneath a macro: the macro doesn't follow, its ring shows the wrong value and the next turn jumps the sound back. Ranges go in Live's MIDI Mappings (Min and Max per mapping): give resonance and feedback the part of their travel that matters, and widen it when the music asks.

— Check the shared knobs once: turn NEXT SONG on one page, go to another and turn it a step. The pick must move a step, not jump back. If it jumps, the README's relative fallback fixes it.

— Keep STORE ALL, MODEL, KEY and the calibration controls off the performance pages.

MACROS

SF CONTROL's six macros, on MIX's MACROS page. Every SF plugin follows one macro (MACRO SRC) by an amount (MACRO AMT, negative turns it the other way); a macro moves the knobs marked M in the module tables, adds to them and is stored per song. A starting plan:

MACRO	FOLLOWED BY	MOVES
1 ENERGY	the group ENERGY plugins, BUILD	every group's energy at once (a negative amount on LOW darkens the bass while the rest lifts)
2 MOOD	PAD, CHORD, VOICE	cutoff, envelope and drive; the voice's SIGNAL and ECHO
3 DENSITY	CLOSED HAT, OPEN HAT, CLAP, PERC	DECAY and DENSITY: the fills
4 PULSE	KICK, RUMBLE	kick DECAY, PUNCH, DRIVE; rumble SIZE, GRIT, DENSITY
5 SPACE	DUB, SPACE returns	delay FEEDBACK and MIX; reverb SIZE, DECAY and MIX
6 FRACTURE	LEAD, BASS	cutoff, envelope, density and drive: smooth to heavy

Start the amounts at 30-50 %. One macro per plugin: a plugin can follow only one.

RETURNS AND POCKET

— A DUB: SF DELAY, DRY off, TIME synced, FEEDBACK mapped to 0-70 % at first, DUCK up so the repeats sit around the kick. INPUT THROW ONLY makes its THROW press the only way in: dub throws.

— B SPACE: SF REVERB, DRY off, DUCK 30-60 % so the tail blooms in the gaps, LO CUT up.

— Sends post-fader: turning a source down stops feeding its effect while the tail rings on. Every track page has the track's two sends (13, 14). No return-to-return sends at first.

— Pocket needs no wiring: SF KICK reports on the bus. SF CONTROL should show BUS SHARED and KICK REPORTING.

— Pulling the kick's volume down leaves the kick playing on the bus: Pocket and RUMBLE keep pumping to a silent kick, which can be the point. For a clean withdrawal, mute the kick lane on the ONE.

SONGS, PATTERNS AND SCENES

SYSTEM	RESPONSIBLE FOR
OXI ONE projects and patterns	Notes, rhythms, gates, accents, the pattern progression.
SF songs (SF CONTROL)	The sound of every SF plugin, called from the E16 (NEXT SONG, GO) and recalled on the bar.
E16 pages	Where your hands are.
Live Session scenes	BRIDGE clips, reference material, rehearsal.
Live Arrangement	Recording the performance.

— The roles stay put for the whole set: a new track changes what they play, not where they are.

— A song and a ONE project are independent: turn NEXT SONG (15) to the next song, press it on the bar before the change, and load any project's patterns on the ONE.

— Rehearse song recall, SF REPEAT and SF BUILD's drop (the cut and the way back, TAIL, DROP DUCK) in your Live before relying on them in a transition.

RECORDING AND REHEARSAL

— Instrument tracks on Monitor In: they play from the ONE whether or not they're armed.

— To record: RESET on SF LIMIT, arm the instrument tracks, REC and ROOM, Automation Arm on, Arrangement record, play. The stereo recording on REC is the set, the one on ROOM the room mics for the mix; the MIDI and automation are for editing and rehearsal.

— To replay recorded MIDI: stop the ONE, set those tracks to Monitor Auto and disarm them (Monitor In ignores a track's own clips).

— Collect All and Save. SF samples are inside the set; DECK tracks are referenced by path, so keep the files where they are.

BUILD ORDER

1. SF SHOW.als (controllers/live): Live's preferences set once, the outputs checked against the interface; saved as a template.
2. CONTROL, 1 KICK, 6 BASS, 7 RUMBLE, LIVE BUS, PA and REC playing. Clock and latency calibrated, a short recording made.
3. 2 CLOSED HAT, 3 OPEN HAT, 4 CLAP and 5 PERC.
4. 8 LEAD, 9 CHORD, 10 PAD and 11 VOICE, one at a time.
5. The returns; check the sends are disabled on LIVE BUS.
6. BRIDGE and DECK.
7. PERF and MIX on the E16 (controllers/e16): both scenes loaded, every page checked against the set's mappings; the macros.
8. Rehearse moving between BRIDGE, DECK and LIVE BUS.
9. Record a whole set before adding anything else.

BEFORE THE FIRST SHOW

— CONTROL shows BUS SHARED and KICK REPORTING; the pump lands on the kick.

— NEXT SONG (15) turned to another song changes nothing; its press (GO) lands the song on the next bar line, every track on the same bar, and a recalled knob is yours the moment you turn it.

— The interface's output knob down: the speakers silent, REC still recording.

— Every E16 hold press (DROP, THROW, ROLL) held for three seconds stays on for the whole hold and goes off on release.

— The shared E16 knobs don't jump: NEXT SONG turned on one page, then a step on another, moves a step. Changing to NRG or a MIX page sends nothing (Live's MIDI input indicator stays dark); a PERF track page selects its track (with SF E16 installed).

— A minute recorded on REC with a deep ROOM EQ cut: the cut isn't in the recording.

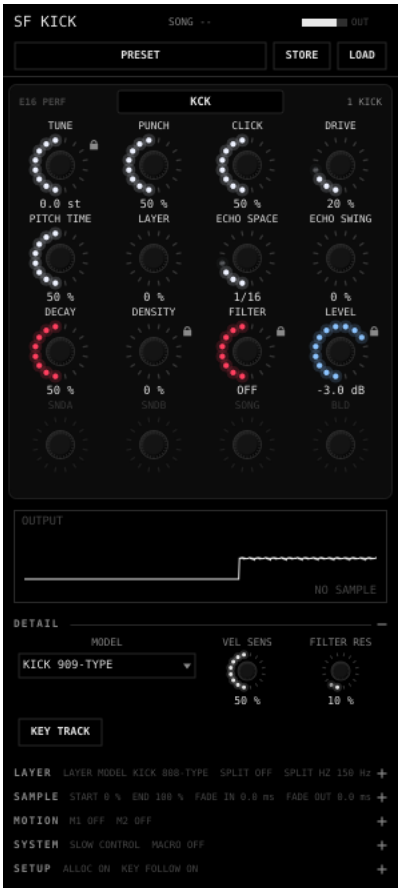
— With SF TERROIR in the set: a capture from ROOM into an empty slot ends SWEEP x3, 30 dB or more over the noise, and the PA stays silent through it.

8 INSTRUMENTS

SF KICK

INSTRUMENT, MIDI TRACK

Kick drum with six synthesised models and a sampler. Every hit is reported to the other SF plugins, which duck around it (Pocket).



USING IT

— Put it on a MIDI track. Any note plays the kick (KEY TRACK off).

— DECAY is your GATE: short and tight for driving sections, long and boomy for breakdowns. RUMBLE and the bass pockets follow the change automatically.

— Ride DENSITY with ECHO SPACE at 1/16 or 1/8D to thicken the kick into a drop. Ghost kicks get smaller pockets.

— High-passing the kick with FILTER hands the sub to the bass and rumble: their pockets close as the kick thins.

— Set the song's KEY on SF CONTROL and the kick tunes itself to the key root (KEY FOLLOW).

MODELS

NAME	SOUND
KICK 909-TYPE	The 909's circuit: a triangle oscillator reset on every hit, swept down fast to about 50 Hz and rounded by a biased diode limiter (a sine with a clipped underside), with a noise-and-pulse attack. PITCH TIME is the 909's TUNE (how fast the sweep falls), PUNCH how high it starts, CLICK the 909's ATTACK. An accent (velocity) hits harder: louder, higher, more click. The default techno kick.
KICK 808-TYPE	The 808's circuit: a 1 ms trigger pulse rings a bridged-T resonator at about 49.5 Hz. For the first few milliseconds its pitch jumps more than an octave (PUNCH), then it sighs down from about 58 Hz over the first 300 ms (deeper on an accent; PITCH TIME its pace). DECAY is the 808's DECAY, CLICK its TONE (the low-pass after it).
KICK SINE	Clean sine with a moderate sweep. Deep and minimal (dub techno).
KICK INDUSTRIAL	Steep sweep, heavy drive and clip. Hard and compressed.
KICK MODULAR	Triangle core with noise FM and a resonant low-pass sweep (DFAM-like).
KICK DUB	Soft, low, slow sweep, filtered dark.
SAMPLE	Plays the dropped one-shot, from START to END. Its pitch and decay are analysed and reported for Pocket (a part cut shorter: its length).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TUNE	-12.0 st to 12.0 st default 0.0 st	Kick pitch in semitones around the model's fundamental (about 44-55 Hz).
DECAY M	0 % to 100 % default 50 %	Body length, x0.25 to x4 of the model's decay. This is the kick's GATE. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
PUNCH M	0 % to 100 % default 50 %	Depth of the pitch sweep. More = harder transient, less = rounder.
CLICK	0 % to 100 % default 50 %	Level of the noise/impulse click on the attack.
DRIVE M	0 % to 100 % default 20 %	Saturation on the kick, added to the model's own.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DENSITY	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
LEVEL	-36.0 dB to 6.0 dB default -3.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PITCH TIME	0 % to 100 % default 50 %	How fast the pitch sweep falls. Short = snappy, long = laser.
VEL SENS	0 % to 100 % default 50 %	How much velocity changes the level. 0 = every hit equal.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
KEY TRACK	ON / OFF default OFF	On: the note number sets the pitch (C1 = TUNE). Off: every note plays the same kick.
FILTER RES	0 % to 100 % default 10 %	Resonance of the output FILTER.

LAYER

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LAYER MODEL	KICK 909-TYPE, KICK 808-TYPE, KICK SINE, KICK INDUSTRIAL, KICK MODULAR, KICK DUB, SAMPLE default KICK 808-TYPE	Second model hit with every note (e.g. 909 click on an 808 body).
LAYER	0 % to 100 % default 0 %	Level of the LAYER MODEL. Main and layer always add up to the level of one kick.
SPLIT	MODEL LOWS 100 % to LAYER LOWS 100 % default OFF	Which kick plays the lows and which the punch and click, crossing over at SPLIT HZ. Centre: OFF, both whole. Left: MODEL's lows, the LAYER MODEL's top; right: the LAYER MODEL's lows (an 808 body under a 909's click). At full SPLIT each band is as loud as its kick alone (the layer at LAYER's level). Through the crossover the lows land about 3 ms after the top at 150 Hz.
SPLIT HZ	50 Hz to 800 Hz default 150 Hz	Where SPLIT crosses over, 50-800 Hz (150 Hz).

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the sample starts playing, from 0 % (its start) to 100 %. Drag the START line on the display or turn this. A note takes it as it is when it starts. Inside the sample it fades in for at least 1 ms, so a cut never clicks.
END	0 % to 100 % default 100 %	Where the sample stops, from START to 100 % (its end). Never before START (1 ms after it at least). Inside the sample it fades out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

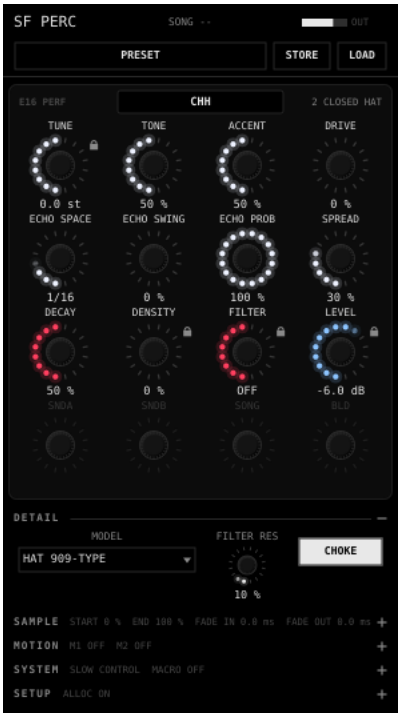
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: with a KEY set on SF CONTROL, the model's fundamental lands on the nearest key root and TUNE moves from there (TUNE 0 = on the root). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF PERC

INSTRUMENT, MIDI TRACK

Hats, cymbals, claps and percussion. Thirteen models including a sampler. Run two or three instances for a full kit.



USING IT

— Program fixed 1/8 hats on the OXI. Bring DENSITY up with ECHO SPACE 1/16 to turn them into 1/16s without changing the pattern; above 50 % they start rolling.

— DECAY opens closed hats into open hats.

— ECHO PROB under 100 % makes the fills irregular; SPREAD throws the ghosts left and right.

— For the real 909 hat sound, load your own samples into the SAMPLE model.

MODELS

NAME	SOUND
HAT 909-TYPE	The 909's way of playing its hats: a recorded-style metal and noise sound, quantised to 6 bits at about 32 kHz as the 909's ROM held it, through its DAC's filter, then the analog VCA (DECAY). The tail is grainy and stops dead, wherever DECAY is.
HAT 808-TYPE	The 808's circuit: its six square oscillators (shared with its cymbal and cowbell) through the 7.1 kHz band-pass, the swing VCA that clips against the envelope (gritty, then fully off), and its resonant high-passes.
HAT 606-TYPE	Thin, noisy, ticky.
RIDE 909-TYPE	Long metallic ride through the 909's 6-bit ROM path.
CRASH	Noise-heavy long crash through the 909's 6-bit ROM path.
CLAP	Three quick noise bursts and a tail.
RIM	Two tuned sines, short.
COWBELL	Two square oscillators through a band-pass, two-stage decay.
CLAVE	Short tuned sine.
SHAKER	Band-passed noise with a soft attack.
METAL FM	Inharmonic two-operator FM hit. Industrial.
MODAL	Six tuned resonators struck by a noise burst. Wood, glass or metal depending on TONE.
SAMPLE	Plays the dropped one-shot from START to END, pitched by TUNE.
SNARE 808-TYPE	The 808's snare: two bridged-T resonators (238 and 476 Hz) and snappy noise. TONE is the snappy.
SNARE 909-TYPE	The 909's snare: two triangle oscillators through its diode limiter, and noise. TONE is the 909's SNAPPY and TONE together: more noise, longer.
TOM 808-TYPE	The 808's tom: a bridged-T at 100 Hz with its attack's pitch jump. TUNE picks low, mid or high; TONE a little noise.
TOM 909-TYPE	The 909's tom: a triangle oscillator swept down onto 110 Hz through its diode limiter. TUNE picks the tom; TONE a noise burst.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch in semitones (-24 to +24).
DECAY M	0 % to 100 % default 50 %	Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
TONE	0 % to 100 % default 50 %	Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
ACCENT	0 % to 100 % default 50 %	How much velocity matters. 0 = all hits equal.
DRIVE	0 % to 100 % default 0 %	Saturation on each hit.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ECHO PROB	0 % to 100 % default 100 %	Chance each ghost note plays. Below 100 % the fills become irregular, like a drummer.
CHOKE	ON / OFF default ON	On: a new hit cuts the previous one (like a closed hat choking an open hat).
SPREAD	0 % to 100 % default 30 %	Pans the DENSITY ghosts alternately left and right.
FILTER RES	0 % to 100 % default 10 %	Resonance of the output FILTER. Higher gives a sharper, more vocal sweep.

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the sample starts playing, from 0 % (its start) to 100 %. Drag the START line on the display or turn this. A note takes it as it is when it starts. Inside the sample it fades in for at least 1 ms, so a cut never clicks.
END	0 % to 100 % default 100 %	Where the sample stops, from START to 100 % (its end). Never before START (1 ms after it at least). Inside the sample it fades out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

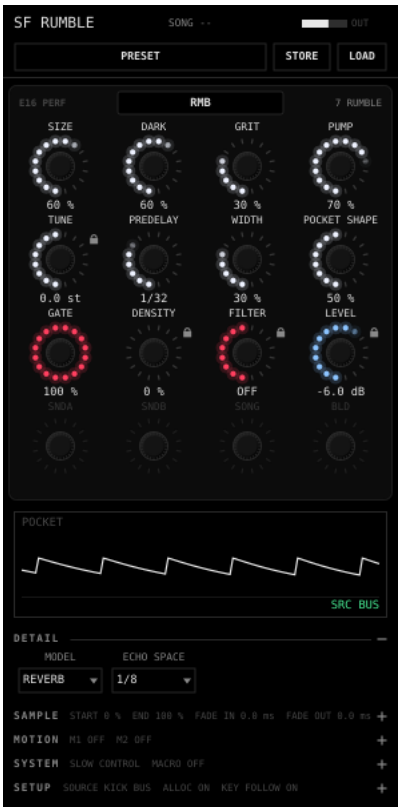
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF RUMBLE

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

The kick-driven low-end bed of hypnotic techno. Needs no notes: it listens to SF KICK, excites a thump with the kick's own pitch sweep, and runs it through a reverb, delay or drone model, then darkens, dirties and ducks it.



USING IT

— Put it on a MIDI track (no clips needed) with SOURCE on KICK BUS.

— SIZE and DARK set the room; GRIT adds the dirt; PUMP decides how hard it ducks out of the kick's way.

— GATE below 100 % cuts the tail before the next kick: tight, rhythmic rumble.

— DENSITY adds extra thumps between the kicks for a rolling bed.

MODELS

NAME	SOUND
REVERB	Thump into a dark reverb. The classic.
DELAY	Thump into a 3/16 feedback delay.
SUB DRONE	Continuous sine at the kick's pitch, ducked by the kick: swells between hits.
DISTORTED	Reverb with heavy drive.
DUB BOOM	Delay and reverb together, softer.
SAMPLE	Plays a dropped rumble one-shot on every kick: up to 0.4 s of it, from START (END and the fades shape it).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIZE M	0 % to 100 % default 60 %	Reverb size / delay feedback.
DARK	0 % to 100 % default 60 %	Low-pass on the bed: 600 Hz down to 70 Hz.
GRIT M	0 % to 100 % default 30 %	Distortion on the bed.
PUMP	0 % to 100 % default 70 %	Pocket depth: how far the bed ducks under each kick.
GATE	0 % to 100 % default 100 %	Cuts the tail after each kick. 100 % = no gate; lower = shorter.
DENSITY M	0 % to 100 % default 0 %	Extra thumps between kicks (offbeat, then 16ths).
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TUNE	-12.0 st to 12.0 st default 0.0 st	Pitch of the thump relative to the kick.
PREDELAY	0, 1/32, 1/16, 1/8 default 1/32	Delay before the reverb: 0, 1/32, 1/16, 1/8.
WIDTH	0 % to 100 % default 30 %	Stereo width of the bed.
POCKET SHAPE	0 % to 100 % default 50 %	Pocket SHAPE (see Pocket chapter).
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/8	Spacing of DENSITY thumps.

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the sample starts playing, from 0 % (its start) to 100 %. Drag the START line on the display or turn this. A note takes it as it is when it starts. Inside the sample it fades in for at least 1 ms, so a cut never clicks.
END	0 % to 100 % default 100 %	Where the sample stops, from START to 100 % (its end). Never before START (1 ms after it at least). Inside the sample it fades out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

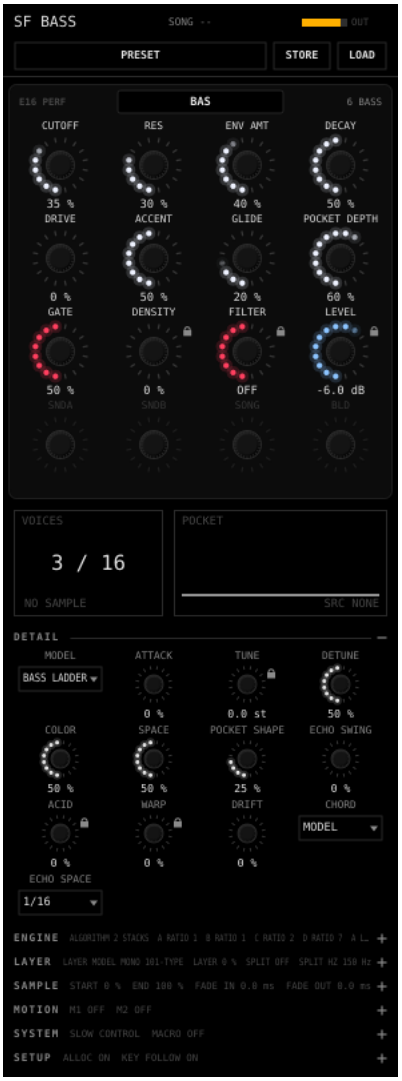
CONTROL	RANGE / DEFAULT	WHAT IT DOES
SOURCE	KICK BUS, MIDI, BOTH default KICK BUS	KICK BUS: triggered by SF KICK. MIDI: by notes on this track. BOTH.
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: with a KEY set on SF CONTROL, the thump lands on the nearest key root. Off: it follows the kick's pitch.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF BASS

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Mono bass synth: ladder, 101, sub-oscillator, acid, FM, sine and Berlin off-beat models. Shared engine with BASS / LEAD / PAD: the same controls on every model.



USING IT

— CUTOFF, RES and ENV AMT are the performance; one note can carry a whole track.

— POCKET DEPTH (default 60 %) keeps the bass out of the kick's way; it follows the kick's pitch and length.

— DENSITY ghost notes fill the gaps between notes but never interrupt a note that is held.

MODELS

NAME	SOUND
BASS LADDER	Two detuned saws and a sub through a Moog-style ladder.
MONO 101-TYPE	Saw + pulse + sub, 4-pole filter.
SUB OSC	Square plus heavy sub-oscillator, soft filter. Deep and round.
ACID 303-TYPE	The 303's circuit: diode ladder filter, envelope, accent sweep and slide (see LEAD).
FM	Two-operator FM bass; COLOR sets the index.
SINE SUB	Pure sine with slight drive. No filter.
OFFBEAT	Square + saw pluck, high resonance. The rubbery Berlin off-beat bass.
SAMPLE	Chromatic sampler from the analysed root note.
FM 4-OP	Four sine operators, A B C D, on one of eight ALGORITHMs (the ENGINE group): each with its RATIO to the note, LEVEL and DECAY, and D's FEEDBACK. COLOR scales every modulator's depth.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUTOFF M	0 % to 100 % default 35 %	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
RES	0 % to 100 % default 30 %	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
ENV AMT M	0 % to 100 % default 40 %	How far the filter envelope opens the filter on each note. Accented notes open further.
GATE	0 % to 100 % default 50 %	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation before the filter. Adds weight at low settings, grit at high.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DECAY	0 % to 100 % default 50 %	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
ATTACK	0 % to 100 % default 0 %	Adds attack time to amp and filter envelopes. Up for swells.
ACCENT	0 % to 100 % default 50 %	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
GLIDE	0 % to 100 % default 20 %	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch offset in semitones (-24 to +24).
DETUNE	0 % to 100 % default 50 %	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
COLOR	0 % to 100 % default 50 %	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
CHORD	MODEL, NONE, MIN7, MIN9, SUS4, MAJ7 ... (9) default MODEL	Notes played per key. MODEL uses the model's own chord (STAB DUB = minor 7, STAB DETROIT = minor 9, ORGAN = minor triad). NONE forces single notes. Or choose MIN7, MIN9, SUS4, MAJ7, MIN, 5TH, OCT.
SPACE	0 % to 100 % default 50 %	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
POCKET DEPTH	0 % to 100 % default 60 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 25 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ACID M	0 % to 100 % default 0 %	One knob for the squelch (the E16's NRG 12, on 8 LEAD): RES towards 95 %, ENV AMT towards 90 %, ACCENT towards full, DRIVE +35 % and CUTOFF +10 %, each from where its own knob is (never down). Made for ACID 303-TYPE; any model takes it. 0 = off.
WARP M	0 % to 100 % default 0 %	One knob for a melting texture (the E16's NRG 14, on 10 PAD): tape wow (up to 35 cents, two slow drifts) and flutter (5 cents), every voice on its own phase and pace so a chord smears; DETUNE and SPACE pushed up too (on GRANULAR: the grains scatter wider). 0 = off.
DRIFT	0 % to 100 % default 0 %	Analog drift. Each voice of a poly model sits a little off tune (up to 5 cents) and off cutoff, always the same voice the same way, as an analog poly's voices do; every voice wanders slowly (up to 4 cents, over 6-14 s). A mono model wanders as its one oscillator. 0 = dead steady.

ENGINE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TABLE	BUILT-IN, SAMPLE default BUILT-IN	WAVETABLE's frames: BUILT-IN (sine, saw, square, vowel) or SAMPLE: eight frames cut from the loaded sample, start to end, and COLOR moves across them. Any recording: a voice, a field recording, a chord. Pitched sounds give clean frames; the note you play sets the pitch.
ALGORITHM	STACK, Y, BRANCH, 2 STACKS, 1 TO 3, STACK + 1, BELL, ORGAN default 2 STACKS	FM 4-OP: how the four operators connect (modulation runs from D down to A). STACK D>C>B>A; Y C and D > B > A; BRANCH D>C, then C and B > A; 2 STACKS B>A and D>C, both heard; 1 TO 3 D into A, B and C; STACK + 1 D>C>B, and A; BELL D>C into A and B; ORGAN all four heard.
A RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 1	Operator A's frequency against the note (0.5 to 16). Whole numbers give harmonic tones, the halves and high ones bells and metal.
B RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 1	Operator B's frequency against the note.
C RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 2	Operator C's frequency against the note.
D RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 7	Operator D's frequency against the note.
A LEVEL	0 % to 100 % default 100 %	Operator A's level: how loud a heard operator is, how deep a modulating one goes (with COLOR).
B LEVEL	0 % to 100 % default 50 %	Operator B's level.
C LEVEL	0 % to 100 % default 60 %	Operator C's level.
D LEVEL	0 % to 100 % default 25 %	Operator D's level.
A DECAY	0 % to 100 % default 70 %	Operator A's own envelope: it falls from the note's start, 0.03 s to 7.5 s to a third (times DECAY's scaling); at the top it holds. A modulator falling makes the tone duller as the note goes on.
B DECAY	0 % to 100 % default 40 %	Operator B's envelope.
C DECAY	0 % to 100 % default 60 %	Operator C's envelope.
D DECAY	0 % to 100 % default 25 %	Operator D's envelope.
FEEDBACK	0 % to 100 % default 20 %	D modulating itself: from a sine toward a saw, then noise.

LAYER

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LAYER MODEL	BASS LADDER, MONO 101-TYPE, SUB OSC, ACID 303-TYPE, FM, SINE SUB ... (9) default MONO 101-TYPE	Second model played with every note, blended by LAYER.
LAYER	0 % to 100 % default 0 %	Blend of the LAYER MODEL under the main model. Use it to put a sub under a stab or an acid line on a bass.
SPLIT	MODEL LOWS 100 % to LAYER LOWS 100 % default OFF	Which layer plays the lows and which the mids and highs, crossing over at SPLIT HZ. Centre: OFF, both play everything (the sound as without it). Left: MODEL owns the lows, the LAYER MODEL the rest; right: the LAYER MODEL owns the lows (a sub under an acid line or a stab). At full SPLIT each band is as loud as its layer alone (the LAYER MODEL at LAYER's level; at 100 % as loud as MODEL), and the lows are mono. With LAYER at 0 the layer is silent, so SPLIT cuts MODEL's lows (right) or its mids and highs (left).
SPLIT HZ	50 Hz to 800 Hz default 150 Hz	Where SPLIT crosses over, 50-800 Hz (150 Hz). Around 100-200 Hz for a sub under a bass line, higher to give a stab the mids and highs only.

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the sample starts playing, from 0 % (its start) to 100 %. Drag the START line on the display or turn this. A SAMPLE note takes it as it is when the note starts (a legato slide plays on); GRANULAR's grains follow it as it moves. Inside the sample it fades in for at least 1 ms, so a cut never clicks.
END	0 % to 100 % default 100 %	Where the sample stops, from START to 100 % (its end). Never before START (1 ms after it at least). Inside the sample it fades out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

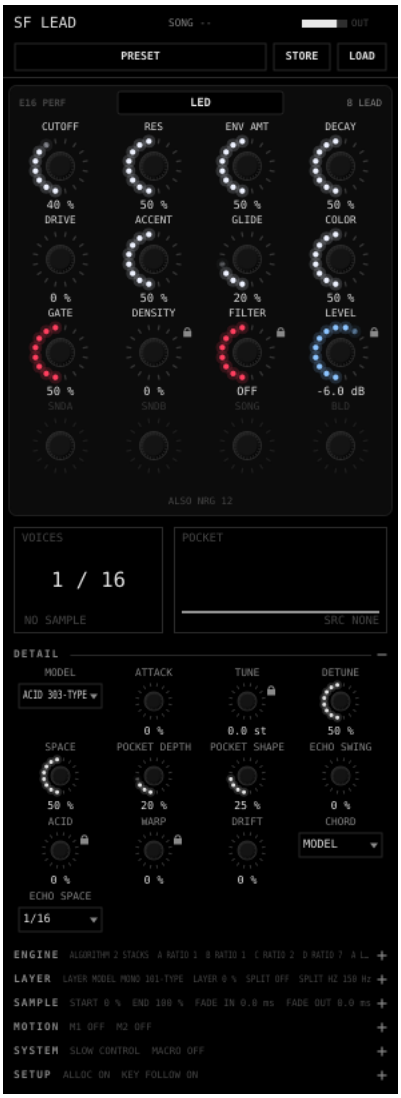
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: follows the KEY set on SF CONTROL (parts written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF LEAD

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Leads, acid lines and chord stabs. Shared engine with BASS / LEAD / PAD: the same controls on every model.



USING IT

— For acid: ACID 303-TYPE, RES high, ENV AMT and DECAY on knobs, accents and slides from the OXI.

— For dub stabs: STAB DUB with GATE short and SF DELAY on a return.

— CHORD overrides the model's chord; NONE gives single notes.

MODELS

NAME	SOUND
ACID 303-TYPE	The 303's circuit: its diode ladder filter (a gentle slope; the bass stays as RES rises), envelope and accent sweep. Velocity above ~108 is an accent: louder, a short envelope, and a sweep that climbs on accents close together (more with RES). Overlapping notes slide without retriggering. ENV AMT widens the sweep both ways, as the 303's ENV MOD. COLOR: saw or the 303's square.
MONO 101-TYPE	Saw + pulse + sub. Classic mono sequence lead.
MS20-TYPE	Driven high-pass + low-pass pair, screaming resonance.
STAB DUB	Minor 7 chord stab, short filter envelope, light ensemble. Send it into SF DELAY.
STAB DETROIT	Minor 9 string/brass stab with unison and ensemble.
ORGAN M1-TYPE	Additive organ with key click and percussion, minor triad.
HOOVER	Detuned pulse-width-modulated stack with a pitch dive and heavy chorus.
FM BELL	Two-operator bell, long release.
SYNC	Hard-sync lead; COLOR and ENV AMT sweep the sync.
SAMPLE	Chromatic sampler from the analysed root note.
FM 4-OP	Four sine operators, A B C D, on one of eight ALGORITHMs (the ENGINE group): each with its RATIO to the note, LEVEL and DECAY, and D's FEEDBACK. COLOR scales every modulator's depth.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUTOFF M	0 % to 100 % default 40 %	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
RES	0 % to 100 % default 50 %	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
ENV AMT M	0 % to 100 % default 50 %	How far the filter envelope opens the filter on each note. Accented notes open further.
GATE	0 % to 100 % default 50 %	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation before the filter. Adds weight at low settings, grit at high.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DECAY	0 % to 100 % default 50 %	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
ATTACK	0 % to 100 % default 0 %	Adds attack time to amp and filter envelopes. Up for swells.
ACCENT	0 % to 100 % default 50 %	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
GLIDE	0 % to 100 % default 20 %	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch offset in semitones (-24 to +24).
DETUNE	0 % to 100 % default 50 %	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
COLOR	0 % to 100 % default 50 %	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
CHORD	MODEL, NONE, MIN7, MIN9, SUS4, MAJ7 ... (9) default MODEL	Notes played per key. MODEL uses the model's own chord (STAB DUB = minor 7, STAB DETROIT = minor 9, ORGAN = minor triad). NONE forces single notes. Or choose MIN7, MIN9, SUS4, MAJ7, MIN, 5TH, OCT.
SPACE	0 % to 100 % default 50 %	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
POCKET DEPTH	0 % to 100 % default 20 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 25 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ACID M	0 % to 100 % default 0 %	One knob for the squelch (the E16's NRG 12, on 8 LEAD): RES towards 95 %, ENV AMT towards 90 %, ACCENT towards full, DRIVE +35 % and CUTOFF +10 %, each from where its own knob is (never down). Made for ACID 303-TYPE; any model takes it. 0 = off.
WARP M	0 % to 100 % default 0 %	One knob for a melting texture (the E16's NRG 14, on 10 PAD): tape wow (up to 35 cents, two slow drifts) and flutter (5 cents), every voice on its own phase and pace so a chord smears; DETUNE and SPACE pushed up too (on GRANULAR: the grains scatter wider). 0 = off.
DRIFT	0 % to 100 % default 0 %	Analog drift. Each voice of a poly model sits a little off tune (up to 5 cents) and off cutoff, always the same voice the same way, as an analog poly's voices do; every voice wanders slowly (up to 4 cents, over 6-14 s). A mono model wanders as its one oscillator. 0 = dead steady.

ENGINE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TABLE	BUILT-IN, SAMPLE default BUILT-IN	WAVETABLE's frames: BUILT-IN (sine, saw, square, vowel) or SAMPLE: eight frames cut from the loaded sample, start to end, and COLOR moves across them. Any recording: a voice, a field recording, a chord. Pitched sounds give clean frames; the note you play sets the pitch.
ALGORITHM	STACK, Y, BRANCH, 2 STACKS, 1 TO 3, STACK + 1, BELL, ORGAN default 2 STACKS	FM 4-OP: how the four operators connect (modulation runs from D down to A). STACK D>C>B>A; Y C and D > B > A; BRANCH D>C, then C and B > A; 2 STACKS B>A and D>C, both heard; 1 TO 3 D into A, B and C; STACK + 1 D>C>B, and A; BELL D>C into A and B; ORGAN all four heard.
A RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 1	Operator A's frequency against the note (0.5 to 16). Whole numbers give harmonic tones, the halves and high ones bells and metal.
B RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 1	Operator B's frequency against the note.
C RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 2	Operator C's frequency against the note.
D RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 7	Operator D's frequency against the note.
A LEVEL	0 % to 100 % default 100 %	Operator A's level: how loud a heard operator is, how deep a modulating one goes (with COLOR).
B LEVEL	0 % to 100 % default 50 %	Operator B's level.
C LEVEL	0 % to 100 % default 60 %	Operator C's level.
D LEVEL	0 % to 100 % default 25 %	Operator D's level.
A DECAY	0 % to 100 % default 70 %	Operator A's own envelope: it falls from the note's start, 0.03 s to 7.5 s to a third (times DECAY's scaling); at the top it holds. A modulator falling makes the tone duller as the note goes on.
B DECAY	0 % to 100 % default 40 %	Operator B's envelope.
C DECAY	0 % to 100 % default 60 %	Operator C's envelope.
D DECAY	0 % to 100 % default 25 %	Operator D's envelope.
FEEDBACK	0 % to 100 % default 20 %	D modulating itself: from a sine toward a saw, then noise.

LAYER

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LAYER MODEL	ACID 303-TYPE, MONO 101-TYPE, MS20-TYPE, STAB DUB, STAB DETROIT, ORGAN M1-TYPE ... (11) default MONO 101-TYPE	Second model played with every note, blended by LAYER.
LAYER	0 % to 100 % default 0 %	Blend of the LAYER MODEL under the main model. Use it to put a sub under a stab or an acid line on a bass.
SPLIT	MODEL LOWS 100 % to LAYER LOWS 100 % default OFF	Which layer plays the lows and which the mids and highs, crossing over at SPLIT HZ. Centre: OFF, both play everything (the sound as without it). Left: MODEL owns the lows, the LAYER MODEL the rest; right: the LAYER MODEL owns the lows (a sub under an acid line or a stab). At full SPLIT each band is as loud as its layer alone (the LAYER MODEL at LAYER's level; at 100 % as loud as MODEL), and the lows are mono. With LAYER at 0 the layer is silent, so SPLIT cuts MODEL's lows (right) or its mids and highs (left).
SPLIT HZ	50 Hz to 800 Hz default 150 Hz	Where SPLIT crosses over, 50-800 Hz (150 Hz). Around 100-200 Hz for a sub under a bass line, higher to give a stab the mids and highs only.

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the sample starts playing, from 0 % (its start) to 100 %. Drag the START line on the display or turn this. A SAMPLE note takes it as it is when the note starts (a legato slide plays on); GRANULAR's grains follow it as it moves. Inside the sample it fades in for at least 1 ms, so a cut never clicks.
END	0 % to 100 % default 100 %	Where the sample stops, from START to 100 % (its end). Never before START (1 ms after it at least). Inside the sample it fades out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

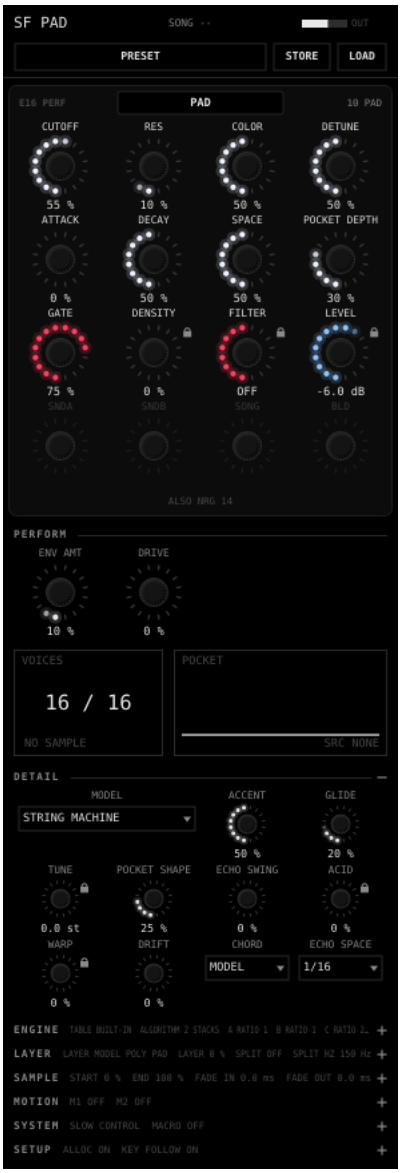
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: follows the KEY set on SF CONTROL (parts written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF PAD

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Pads, drones and textures. Polyphonic, slow envelopes. Shared engine with BASS / LEAD / PAD: the same controls on every model.



USING IT

— GATE high keeps chords ringing; SLOW mode on CUTOFF gives long, even swells.

— GRANULAR and SAMPLE use a dropped file. Field recordings work well in GRANULAR: set START and END around the part you want the grains from, and move them while a chord holds to wander through it.

— POCKET DEPTH lets the pad breathe with the kick without a sidechain.

MODELS

NAME	SOUND
STRING MACHINE	Four-voice unison saws plus an octave, deep ensemble.
POLY PAD	Saw + pulse + sub, slow filter envelope, chorus (Juno-like).
WAVETABLE	Morphs sine > saw > square > vocal with COLOR and a slow drift.
GRANULAR	Grains from the loaded sample's START..END (or a built-in chord bed). COLOR = grain size, DETUNE = scatter. Move START and END while a note holds and the grains follow.
DUB WASH	Minor 7 chord, slow filter, chorus. The dub chord bed.
TAPE LOOP	Soft triangle/saw with wow, flutter, saturation and hiss.
NOISE BED	Noise through a resonant band-pass at the note: wind, air.
SAMPLE	Chromatic sampler from the analysed root note.
FM 4-OP	Four sine operators, A B C D, on one of eight ALGORITHMs (the ENGINE group): each with its RATIO to the note, LEVEL and DECAY, and D's FEEDBACK. COLOR scales every modulator's depth.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUTOFF M	0 % to 100 % default 55 %	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
RES	0 % to 100 % default 10 %	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
ENV AMT M	0 % to 100 % default 10 %	How far the filter envelope opens the filter on each note. Accented notes open further.
GATE	0 % to 100 % default 75 %	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation before the filter. Adds weight at low settings, grit at high.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DECAY	0 % to 100 % default 50 %	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
ATTACK	0 % to 100 % default 0 %	Adds attack time to amp and filter envelopes. Up for swells.
ACCENT	0 % to 100 % default 50 %	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
GLIDE	0 % to 100 % default 20 %	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch offset in semitones (-24 to +24).
DETUNE	0 % to 100 % default 50 %	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
COLOR	0 % to 100 % default 50 %	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
CHORD	MODEL, NONE, MIN7, MIN9, SUS4, MAJ7 ... (9) default MODEL	Notes played per key. MODEL uses the model's own chord (STAB DUB = minor 7, STAB DETROIT = minor 9, ORGAN = minor triad). NONE forces single notes. Or choose MIN7, MIN9, SUS4, MAJ7, MIN, 5TH, OCT.
SPACE	0 % to 100 % default 50 %	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
POCKET DEPTH	0 % to 100 % default 30 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 25 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
ACID M	0 % to 100 % default 0 %	One knob for the squelch (the E16's NRG 12, on 8 LEAD): RES towards 95 %, ENV AMT towards 90 %, ACCENT towards full, DRIVE +35 % and CUTOFF +10 %, each from where its own knob is (never down). Made for ACID 303-TYPE; any model takes it. 0 = off.
WARP M	0 % to 100 % default 0 %	One knob for a melting texture (the E16's NRG 14, on 10 PAD): tape wow (up to 35 cents, two slow drifts) and flutter (5 cents), every voice on its own phase and pace so a chord smears; DETUNE and SPACE pushed up too (on GRANULAR: the grains scatter wider). 0 = off.
DRIFT	0 % to 100 % default 0 %	Analog drift. Each voice of a poly model sits a little off tune (up to 5 cents) and off cutoff, always the same voice the same way, as an analog poly's voices do; every voice wanders slowly (up to 4 cents, over 6-14 s). A mono model wanders as its one oscillator. 0 = dead steady.

ENGINE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TABLE	BUILT-IN, SAMPLE default BUILT-IN	WAVETABLE's frames: BUILT-IN (sine, saw, square, vowel) or SAMPLE: eight frames cut from the loaded sample, start to end, and COLOR moves across them. Any recording: a voice, a field recording, a chord. Pitched sounds give clean frames; the note you play sets the pitch.
ALGORITHM	STACK, Y, BRANCH, 2 STACKS, 1 TO 3, STACK + 1, BELL, ORGAN default 2 STACKS	FM 4-OP: how the four operators connect (modulation runs from D down to A). STACK D>C>B>A; Y C and D > B > A; BRANCH D>C, then C and B > A; 2 STACKS B>A and D>C, both heard; 1 TO 3 D into A, B and C; STACK + 1 D>C>B, and A; BELL D>C into A and B; ORGAN all four heard.
A RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 1	Operator A's frequency against the note (0.5 to 16). Whole numbers give harmonic tones, the halves and high ones bells and metal.
B RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 1	Operator B's frequency against the note.
C RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 2	Operator C's frequency against the note.
D RATIO	0.5, 1, 1.5, 2, 2.5, 3 ... (18) default 7	Operator D's frequency against the note.
A LEVEL	0 % to 100 % default 100 %	Operator A's level: how loud a heard operator is, how deep a modulating one goes (with COLOR).
B LEVEL	0 % to 100 % default 50 %	Operator B's level.
C LEVEL	0 % to 100 % default 60 %	Operator C's level.
D LEVEL	0 % to 100 % default 25 %	Operator D's level.
A DECAY	0 % to 100 % default 70 %	Operator A's own envelope: it falls from the note's start, 0.03 s to 7.5 s to a third (times DECAY's scaling); at the top it holds. A modulator falling makes the tone duller as the note goes on.
B DECAY	0 % to 100 % default 40 %	Operator B's envelope.
C DECAY	0 % to 100 % default 60 %	Operator C's envelope.
D DECAY	0 % to 100 % default 25 %	Operator D's envelope.
FEEDBACK	0 % to 100 % default 20 %	D modulating itself: from a sine toward a saw, then noise.

LAYER

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LAYER MODEL	STRING MACHINE, POLY PAD, WAVETABLE, GRANULAR, DUB WASH, TAPE LOOP ... (9) default POLY PAD	Second model played with every note, blended by LAYER.
LAYER	0 % to 100 % default 0 %	Blend of the LAYER MODEL under the main model. Use it to put a sub under a stab or an acid line on a bass.
SPLIT	MODEL LOWS 100 % to LAYER LOWS 100 % default OFF	Which layer plays the lows and which the mids and highs, crossing over at SPLIT HZ. Centre: OFF, both play everything (the sound as without it). Left: MODEL owns the lows, the LAYER MODEL the rest; right: the LAYER MODEL owns the lows (a sub under an acid line or a stab). At full SPLIT each band is as loud as its layer alone (the LAYER MODEL at LAYER's level; at 100 % as loud as MODEL), and the lows are mono. With LAYER at 0 the layer is silent, so SPLIT cuts MODEL's lows (right) or its mids and highs (left).
SPLIT HZ	50 Hz to 800 Hz default 150 Hz	Where SPLIT crosses over, 50-800 Hz (150 Hz). Around 100-200 Hz for a sub under a bass line, higher to give a stab the mids and highs only.

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the sample starts playing, from 0 % (its start) to 100 %. Drag the START line on the display or turn this. A SAMPLE note takes it as it is when the note starts (a legato slide plays on); GRANULAR's grains follow it as it moves. Inside the sample it fades in for at least 1 ms, so a cut never clicks.
END	0 % to 100 % default 100 %	Where the sample stops, from START to 100 % (its end). Never before START (1 ms after it at least). Inside the sample it fades out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

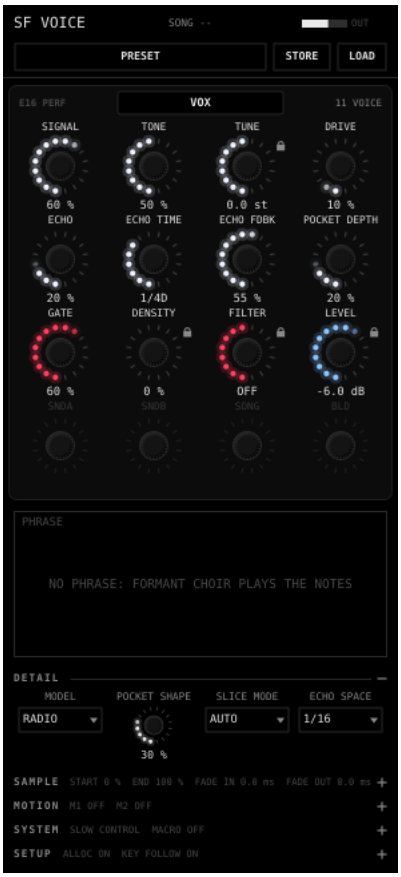
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: follows the KEY set on SF CONTROL (parts written in C play in the song's key). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF VOICE

INSTRUMENT, MIDI TRACK LATENCY 512 SAMPLES AT 48 KHZ

Spoken word and vocal phrases. Drop a phrase: it is sliced, and C3 plays slice 1, C#3 slice 2 and so on; keys below C3 play the whole phrase. Models treat it as radio, vocoder, robot or choir. Built-in dub echo. drosaka Voice stays separate.



USING IT

— Drop a phrase on the window (or LOAD). Markers show the slices and their keys. Drag START and END on the phrase to slice only part of it.

— RADIO with SIGNAL low is a lost transmission; ride SIGNAL up to bring the voice in clear.

— ECHO is a send into a long dub echo (ECHO TIME, ECHO FDBK).

— With no phrase loaded, notes play a formant choir (TONE morphs the vowel).

MODELS

NAME	SOUND
SAMPLE	Plays the slices clean.
RADIO	Band-limited, driven, noisy: CB / transmission. SIGNAL = clarity.
VOCODER	16-band vocoder, saw carrier at the TONE pitch.
ROBOT	Vocoder with a pulse carrier at 8 kHz / 8-bit.
CHOIR	Vocoder with a minor-chord carrier.
STUTTER	Loops the start of each slice; TONE sets the loop length.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIGNAL M	0 % to 100 % default 60 %	Clarity. RADIO: from lost to clear. Vocoders: sibilance and high-band brightness.
TONE	0 % to 100 % default 50 %	Model-specific: carrier pitch C1-C4 (vocoder, robot, choir), loop length (STUTTER), vowel (choir synth).
GATE	0 % to 100 % default 60 %	Slice length. Low values clip each slice short.
DENSITY M	0 % to 100 % default 0 %	Re-triggers the slice as ghost repeats (stutter fills).
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
ECHO M	0 % to 100 % default 20 %	Send into the built-in dub echo.
DRIVE	0 % to 100 % default 10 %	Saturation on the voice.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SLICE MODE	AUTO, EQUAL 8, EQUAL 16, WHOLE default AUTO	AUTO (detected onsets), EQUAL 8, EQUAL 16, WHOLE (every key plays the whole phrase). Slices are cut from the part between START and END.
TUNE	-12.0 st to 12.0 st default 0.0 st	Playback pitch (varispeed) in semitones.
ECHO TIME	1 BAR, 1/2, 1/4, 1/4D, 1/4T, 1/8 ... (9) default 1/4D	Echo time, synced (1 BAR to 1/16). Default 1/4 dotted.
ECHO FDBK	0 % to 100 % default 55 %	Echo feedback.
POCKET DEPTH	0 % to 100 % default 20 %	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
POCKET SHAPE	0 % to 100 % default 30 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.

SAMPLE

CONTROL	RANGE / DEFAULT	WHAT IT DOES
START	0 % to 100 % default 0 %	Where the phrase starts, 0-100 %. The slices are cut from START to END: AUTO's first slice starts at START (or at an onset within 40 ms after it), EQUAL 8 / 16 divide START..END, keys below C3 and WHOLE play START..END. Drag the START line on the phrase or turn this.
END	0 % to 100 % default 100 %	Where the phrase ends, from START to 100 %. The last slice ends here, fading out for at least 1 ms.
FADE IN	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade after START, 0 to 2000 ms of the sample's own time (an octave up it passes in half the time). Longer than the part, the two fades are scaled to fit it.
FADE OUT	0.0 ms to 2.00 s default 0.0 ms	Raised-cosine fade before END, 0 to 2000 ms of the sample's own time. The instrument's own envelope still shapes the note on top.

SETUP

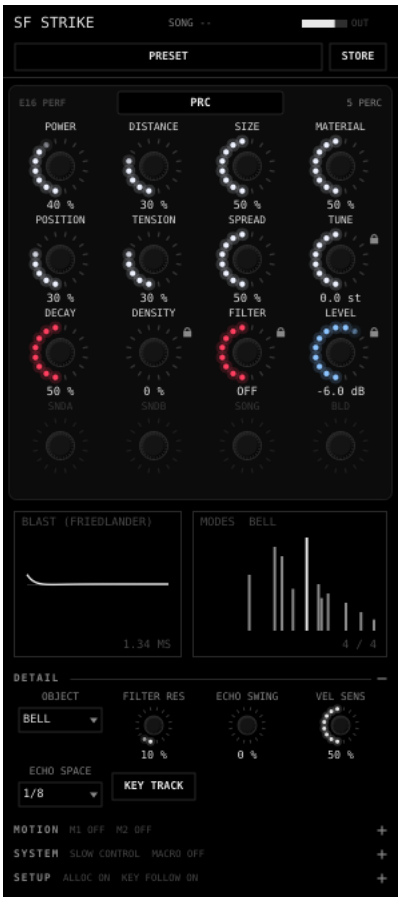
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: the carriers (VOCODER, ROBOT, CHOIR) and the choir played with no phrase loaded follow the KEY set on SF CONTROL. A loaded phrase keeps its own pitch (TUNE moves it). Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF STRIKE

INSTRUMENT, MIDI TRACK

Blast-excited resonant objects. A pressure wave from a charge of POWER at DISTANCE strikes a bell, plate, bar, tube, membrane or gong, modelled as up to 16 resonant modes. Industrial impacts, distant detonations, metal that rings for bars.



USING IT

— Put it on a MIDI track. Any note strikes the object (KEY TRACK on: the note sets the pitch, C3 = TUNE).

— POWER is the charge: low is a tap, high a blast. Above about 70 % the object FRACTURES: its modes split and detune, debris rattles and it distorts. The display shows FRACTURE in amber.

— DISTANCE moves the blast away: quieter, longer and darker, exciting the low modes. Close and small is a sharp crack; far and big is a deep push.

— SIZE sets the pitch (bigger = lower). MATERIAL sets how long the high modes ring (wood to glass). DECAY sets the overall ring.

— DENSITY adds ghost strikes like on the other instruments; each ghost is a smaller charge.

— With a KEY set on SF CONTROL the object's main partial lands on the key root; SIZE then picks the octave.

OBJECTS

NAME	SOUND
BELL	Church-bell partials (hum, prime, tierce, quint, nominal). Long ring; the nominal is the pitch you hear.
PLATE	Sheet metal: dense pairs of close modes that beat. Thunder sheet, industrial clang.
BAR	Free bar (anvil, xylophone ratios). Short and pure.
TUBE	Pipe or tubular bell. Bright and long.
MEMBRANE	Drum head (circular membrane ratios). Short, thuddy.
GONG	Sixteen dense modes, the longest ring, and the strongest pitch glide after a hard hit (TENSION).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
POWER M	0 % to 100 % default 40 %	Size of the charge. Velocity scales it (VEL SENS). Above about 70 % the object fractures.
DISTANCE	0 % to 100 % default 30 %	Stand-off. Further = quieter, longer, darker blast that excites fewer high modes.
SIZE	0 % to 100 % default 50 %	Object size, which sets the pitch: the fundamental goes from about 1.4 kHz (small) to 45 Hz (large).
MATERIAL	0 % to 100 % default 50 %	Wood (0) to glass (100 %): how long the high modes ring compared with the low ones.
DECAY M	0 % to 100 % default 50 %	Ring time, 0.15 to 9 s, times the object's own (a gong rings longer than a bar). Four strikes ring at once; a fifth fades the quietest out over 5 ms.
DENSITY M	0 % to 100 % default 0 %	Ghost notes. 0-50 %: one echo of every note fades in one ECHO SPACE later (1/8 hats become 1/16 hats with ECHO SPACE at 1/16). 50-100 %: two more ghosts at half that spacing (1/32 rolls). Each ghost is quieter, darker and shorter than the note before it. Ghosts are played softer and also faded in after the instrument's velocity response, so at the first notch they are 30 dB down: a fill really rises from silence. A quiet ghost never cuts the ringing kick or chokes the previous hit.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
LEVEL	-36.0 dB to 6.0 dB default -6.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
POSITION	0 % to 100 % default 30 %	Strike point. Striking near a node mutes the modes that have one there, so it changes the colour.
SPREAD	0 % to 100 % default 50 %	Spreads the modes across the stereo field.
TENSION	0 % to 100 % default 30 %	How sharp a hard hit sounds before it settles (the pitch glides down after the blast).
TUNE	-24.0 st to 24.0 st default 0.0 st	Pitch in semitones (-24 to +24).
KEY TRACK	ON / OFF default OFF	On: the note number sets the pitch (C3 = TUNE). Off: every note strikes the same pitch.
FILTER RES	0 % to 100 % default 10 %	Resonance of the output FILTER.
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/8	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO SWING	0 % to 100 % default 0 %	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
VEL SENS	0 % to 100 % default 50 %	How much velocity scales the charge. 0 = every strike the same.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.
KEY FOLLOW	ON / OFF default ON	On: with a KEY set on SF CONTROL, the main partial lands on the nearest key root and TUNE moves from there. Off: ignores KEY.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DECK

INSTRUMENT, MIDI TRACK

Two DJ decks inside the live set. Tracklist with background analysis (tempo, beat grid, loudness), tempo sync to Live with key lock, starts on phrase boundaries, hot cues from MIDI, slip loop roll, per-deck isolator and filter, crossfader, CUE output.



USING IT

— Drop tracks on the window. Each is analysed in the background (ANALYSING, then BPM and LUFS).

— Click A or B on a row to load it into a deck. X removes it from the list.

— PLAY A starts the deck on the next QUANT boundary (default 4 bars), locked to Live's tempo. PLAY off fades it out. Starts, hot-cue jumps and GATE follow the groove: with SF KICK in the set they land where its kicks do, which with a sequencer clocked by Live is a little after Live's beat (the ONE: 30-45 ms).

— Hot cues: C1-D#1 jump deck A to cue 1-4 on the next beat, C2-D#2 deck B. Default cues at beats 0, 32, 64, 96.

— Headphones: make an audio track with Audio From = the DECK track, choose CUE, route it to your headphone outputs, and turn CUE A or CUE B on.

— Every deck start is written to Documents/SF Live Suite/tracklog.txt.

— Decks follow Live's tempo. A plugin cannot change Live's tempo: use Live's tempo automation for tempo moves.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
XFADE	0 % to 100 % default 0 %	Crossfader, constant power. 0 = deck A, 100 % = deck B.
FILTER A	LP 100 to HP 100 default OFF	Deck A bipolar DJ filter.
FILTER B	LP 100 to HP 100 default OFF	Deck B bipolar DJ filter.
LOW A	0 % to 100 % default 75 %	Deck A low band: 0 = kill, 75 % = 0 dB, 100 % = +6 dB.
LOW B	0 % to 100 % default 75 %	Deck B low band.
ROLL	OFF, 1/4, 1/8, 1/16, 1/32 default OFF	Slip loop roll on the playing decks: 1/4 to 1/32. The track keeps running underneath; OFF returns to where it would have been.
GATE	0 % to 100 % default 0 %	1/16 trance gate depth on the deck mix, on the groove's sixteenths.
LEVEL	-36.0 dB to 6.0 dB default -3.0 dB	Output level. Default is set so the module sits sensibly in a mix.
PLAY A	ON / OFF default OFF	Start (on the next QUANT boundary) / stop deck A.
PLAY B	ON / OFF default OFF	Start / stop deck B.
CUE A	ON / OFF default OFF	Send deck A to the CUE output (pre-crossfader).
CUE B	ON / OFF default OFF	Send deck B to the CUE output.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
MID A	0 % to 100 % default 75 %	Deck A mid band.
HIGH A	0 % to 100 % default 75 %	Deck A high band.
MID B	0 % to 100 % default 75 %	Deck B mid band.
HIGH B	0 % to 100 % default 75 %	Deck B high band.
FILTER RES	0 % to 100 % default 20 %	Resonance of both deck filters.

SETUP

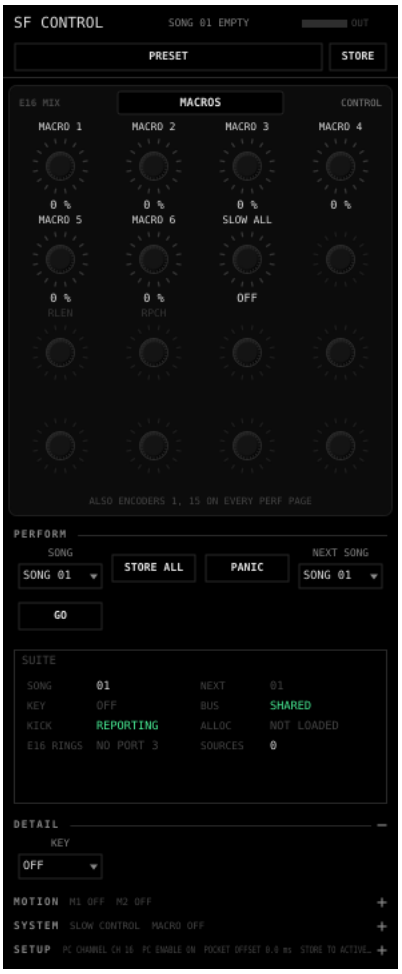
CONTROL	RANGE / DEFAULT	WHAT IT DOES
QUANT	1 BEAT, 1 BAR, 4 BAR, 8 BAR, 16 BAR default 4 BAR	Where PLAY starts: next beat, bar, 4, 8 or 16 bars.
KEY LOCK	ON / OFF default ON	On: time-stretch keeps the track's pitch when its tempo changes. Off: varispeed.
TRIM TARGET	-16.0 LUFS to -6.0 LUFS default -10.0 LUFS	Every track is trimmed to this loudness (LUFS) so decks match.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF CONTROL

INSTRUMENT, MIDI TRACK

The suite's brain. Six macros every SF plugin can follow, song select (NEXT SONG and GO from the E16, SONG, Program Change), STORE ALL and STORE TO, and SLOW ALL. Makes no sound.



USING IT

— Put it on its own MIDI track (CONTROL in SF SHOW: port A, channel 16).

— From the E16 [Nov 8]: turn NEXT SONG (encoder 15 of every PERF page) to pick the song, then press it (GO): the song lands on the next bar line. NEXT SONG alone changes nothing, so a knob knocked in passing never changes the song; the display shows NEXT in amber, PRESS GO, until it is called. GO on the song already in force does nothing.

— Program Change on channel 16 selects the song too, for the AU: Live 12.4 sends a VST3 none (the show set's SF CONTROL is a VST3, so its songs come from the E16).

— Every plugin changes on the same bar line, whatever order Live processes the tracks in. A Program Change up to a sixteenth after a bar line counts for that bar (a sequencer sends it on its downbeat, and one clocked by Live plays a little after Live's: the OXI One 40 ms on Nov 7), a turn of SONG up to 30 ms after it; later in the bar, for the next one.

— To save a song: set every plugin and press STORE ALL. It goes into the song in force (SONG).

— To save into another song, choose it in STORE TO first, then press STORE ALL. That song isn't recalled, so nothing you hear changes; it becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG. If a song change is waiting for its bar, that change still lands and stays the song in force.

— The display shows macro levels, the song, the key, whether the bus is shared, SF KICK is reporting and SF ALLOC is running, and the E16 RINGS port.

— KEY: write parts in C and set each song's key here. Every pitched SF instrument follows; KICK, RUMBLE and STRIKE tune themselves to the key root.

— PANIC, held in this window: every SF plugin fades out over 5 ms, clears every voice, tail and delay line, and stays silent while you hold it. In a show the emergency stop is the audio interface's output knob, which also silences Live's own clips.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
MACRO 1	0 % to 100 % default 0 %	Macro 1. Any SF plugin with MACRO SRC = MACRO 1 follows it.
MACRO 2	0 % to 100 % default 0 %	Macro 2.
MACRO 3	0 % to 100 % default 0 %	Macro 3.
MACRO 4	0 % to 100 % default 0 %	Macro 4.
MACRO 5	0 % to 100 % default 0 %	Macro 5.
MACRO 6	0 % to 100 % default 0 %	Macro 6.
SONG	SONG 01, SONG 02, SONG 03, SONG 04, SONG 05, SONG 06 ... (16) default SONG 01	Selects the song (01-16). Every SF plugin recalls its stored values on the next bar. A recalled knob is yours again the moment you turn it.
STORE ALL	HOLD	Stores every SF plugin's current values into the song STORE TO names: ACTIVE SONG, the song in force, unless you chose another.
PANIC	HOLD	Hold: every SF plugin fades out over 5 ms, clears every voice, tail and delay line and stays silent (no input, no new notes) until you let go; every header shows PANIC in red. Release: back in over 5 ms; DECK keeps its place in time. A set never reopens with PANIC held.
NEXT SONG	SONG 01, SONG 02, SONG 03, SONG 04, SONG 05, SONG 06 ... (16) default SONG 01	Picks the song GO calls (SONG 01-16). Turning it changes nothing you hear; it follows the song in force whenever that changes (GO, SONG, a Program Change, STORE TO), so a pick not yet called gives way. On the E16: encoder 15 of every PERF page.
GO	HOLD	Press: calls the song NEXT SONG picked, on the next bar line (at once with the transport stopped), like a turn of SONG. Nothing when it is already the song in force. On the E16: encoder 15's press.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
KEY	OFF, C, C#, D, D#, E ... (13) default OFF	Key of the song: OFF, C ... B. Parts written in C play in this key on every SF instrument with KEY FOLLOW on (nearest transposition, -6 to +5 semitones). Stored per song, it changes on the song's bar; sounding notes follow. Only while SF CONTROL runs.
SLOW ALL	OFF, 1 BAR, 2 BAR, 4 BAR, 8 BAR, 16 BAR ... (9) default OFF	How long your next knob moves take (OFF, 1-128 BARS) in every SF plugin whose SLOW is CONTROL. A move keeps its length: turning SLOW ALL never touches a travelling knob. A move in the same audio buffer as a SLOW ALL change takes the length from before it, in every plugin alike (with the transport stopped, the new length counts at once). Song recalls follow it. Only while SF CONTROL runs.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PC CHANNEL	CH 1, CH 2, CH 3, CH 4, CH 5, CH 6 ... (16) default CH 16	MIDI channel that Program Change is read from (default 16).
PC ENABLE	ON / OFF default ON	Turns Program Change song select on or off.
POCKET OFFSET	-20.0 ms to 20 ms default 0.0 ms	Timing calibration for Pocket and RUMBLE, -20 to +20 ms (+ = later). Set once, by ear, if the pump lands early or late in your Live setup; leave at 0 otherwise.
STORE TO	ACTIVE SONG, SONG 01, SONG 02, SONG 03, SONG 04, SONG 05 ... (17) default ACTIVE SONG	Where STORE ALL stores: ACTIVE SONG (the song in force) or SONG 01-16. Another song isn't recalled: what plays now goes in, that song becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG.
E16 RINGS	OFF, PORT 1, PORT 2, PORT 3 default PORT 3	E16 port for GLIDE RINGS (OFF, PORT 1-3): in the PERF and MIX scenes a travelling knob's ring turns green. Display: PORT 3 in green once found.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

9 EFFECTS

SF STRIP

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

The suite's performance row for any track. Put it after any Live instrument, sampler, drum rack or hardware input: the track gets DENSITY, GATE, FILTER, DRIVE, POCKET and a SPACE send, joins ALLOC, and follows SLOW, macros and songs like every SF plugin.



USING IT

— Last device on the track (after the instrument and anything else).

— Live sends no MIDI to audio effects, so STRIP works on the hits it hears. GATE closes each detected hit after a set length (staccato to held); DENSITY adds darker, quieter echo ghosts between hits.

— Set ROLE so ALLOC knows the track's priority (drum rack = PERC, a bass synth = BASS).

— THROW sends the track into its own SPACE (dub echo or reverb) while held.

— On a long, sustained sound GATE and DENSITY act on its attacks only; SENSITIVITY sets how small an attack counts as a hit.

MODELS

NAME	SOUND
Chain	GATE > DENSITY > DRIVE > FILTER > SPACE > POCKET > ALLOC > LEVEL (fixed order).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DENSITY M	0 % to 100 % default 0 %	Ghost echoes of each hit: 0-50 % one echo one ECHO SPACE later, 50-100 % two more at half spacing. Each darker and quieter.
GATE M	0 % to 100 % default 100 %	GATE MODE HITS: how long each detected hit stays open, from 1/64 beat at 0 % to 2 beats; 97 % and up = never closes. PATTERN: how long each pattern step stays open.
FILTER	LP 100 to HP 100 default OFF	Bipolar DJ filter on the output. Centre is open. Left sweeps a 24 dB/oct low-pass down to 60 Hz; right sweeps a high-pass up to 12 kHz. Resonance is level-compensated, so a full sweep never jumps in volume.
DRIVE M	0 % to 100 % default 0 %	Saturation, 0 to +30 dB into the shaper (CHARACTER sets tape to fuzz), 4x oversampled, fading in from clean.
POCKET	0 % to 100 % default 0 %	Pocket depth: ducks the track around SF KICK.
SPACE M	0 % to 100 % default 0 %	Send into the strip's own SPACE (dub echo or reverb). The reverb rings out when the send closes.
THROW	HOLD	Hold: full send into SPACE.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ECHO SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
ECHO PROB	0 % to 100 % default 100 %	Chance each hit gets its echoes.
DARKEN	0 % to 100 % default 40 %	How much darker each generation of echoes is.
GATE MODE	HITS, PATTERN default HITS	HITS (closes each detected hit) or PATTERN (rhythmic gate).
PATTERN	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP ... (12) default 8THS	Gate pattern for PATTERN mode.
RATE	1/8, 1/16, 1/32 default 1/16	Step size for PATTERN mode: 1/8, 1/16, 1/32. The steps sit on the groove's grid (where SF KICK's kicks land; Live's grid with no SF KICK).
CHARACTER	0 % to 100 % default 0 %	Drive character: tape (0) to fuzz (100 %).
FILTER RES	0 % to 100 % default 20 %	Resonance of the output FILTER. Higher gives a sharper, more vocal sweep.
POCKET SHAPE	0 % to 100 % default 30 %	One knob from transparent to pumping. Left: a narrow cut that follows the kick's pitch sweep (felt more than heard). Middle: the low band ducks for the kick's length. Right: full-band pump.
SPACE TYPE	REVERB, DUB ECHO default DUB ECHO	REVERB or DUB ECHO (ping-pong, 1/8 to 3/16).
SPACE SIZE	0 % to 100 % default 50 %	Reverb size and decay / echo time and feedback.
SENSITIVITY	0 % to 100 % default 50 %	How small an attack still counts as a hit (GATE HITS and ECHO PROB).

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ROLE	KICK, BASS, VOICE, LEAD, PERC, RUMBLE ... (9) default OTHER	Priority in ALLOC and the name shown (STRIP BASS, STRIP PERC ...).
ALLOC	ON / OFF default ON	Takes part in SF ALLOC (the mix assigner): this module reports its energy per band and accepts cuts where it collides with something more important. Off = never cut.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF FILTER

AUDIO EFFECT

Bipolar performance filter with drive, envelope follow and a throw.



USING IT

— One knob from low-pass through open to high-pass.

— Hold THROW for a timed sweep to the extreme (THROW TIME); release and it returns.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
FILTER M	LP 100 to HP 100 default OFF	Bipolar 24 dB/oct filter. Centre open, left low-pass, right high-pass.
RES	0 % to 100 % default 20 %	Resonance, level-compensated.
DRIVE	0 % to 100 % default 0 %	Saturation before the filter, 2x oversampled. Fades in from exactly clean, so the first notch never jumps the level.
ENV AMT	-100 % to 100 % default 0 %	Envelope follow: the input's level pushes the cutoff up (+) or down (-).
ENV SPEED	0 % to 100 % default 50 %	Speed of the envelope follower.
THROW	HOLD	Hold: the filter travels to the extreme over THROW TIME. Release: back in a quarter of that.
THROW TIME	4 BAR, 2 BAR, 1 BAR, 1/2, 1/4, 1/4D ... (11) default 1/4	Length of the throw: 4 BAR to 1/16.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
THROW TO	HP, LP default HP	Direction of the throw: HP or LP.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF GATE

AUDIO EFFECT

Rhythmic gate with 16 patterns, morphing between two of them.



USING IT

— The tool that turns a held note into a rhythm.

— Set PATTERN A and B, then ride MORPH between them; DEPTH sets how hard the gate cuts.

— The steps follow the groove: with SF KICK in the set they sit where its kicks land against Live's beat (a sequencer clocked by Live plays a little after it: the ONE 30-45 ms), so the gate opens with the notes, not ahead of them. With no SF KICK, Live's grid.

REFERENCE

NAME	SOUND
Patterns	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP, SKIP 1, SKIP 2, STAB, PUMP, BROKEN, CLAVE, 3-3-2, HALF, SPARSE, DOTTED.

PERFORM

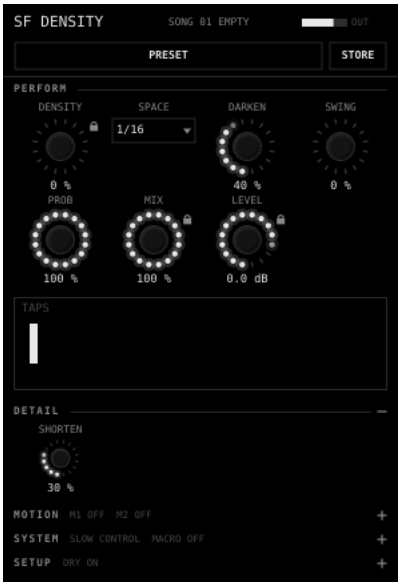
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DEPTH M	0 % to 100 % default 0 %	How far the gate closes. 0 = no effect.
LENGTH	0 % to 100 % default 50 %	How long each step stays open.
SHAPE	0 % to 100 % default 30 %	Edges: hard (clicky) to soft (swelling).
SWING	0 % to 100 % default 0 %	Delays every second step.
PATTERN A	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP ... (16) default 8THS	First pattern.
PATTERN B	FULL, 8THS, OFFBEAT, OFF 8THS, TRESILLO, GALLOP ... (16) default OFFBEAT	Second pattern.
MORPH	0 % to 100 % default 0 %	Blend from PATTERN A to PATTERN B. In between, steps only in one pattern play partly open.
RATE	1/8, 1/16, 1/32, 1/16T default 1/16	Step size: 1/8, 1/16, 1/32, 1/16T.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DENSITY

AUDIO EFFECT

The send-delay hat trick as a plugin: fills sparse rhythms with ghost echoes, each generation darker and quieter.



USING IT

— Put it on hats or percussion (insert, or a return with DRY off).

— DENSITY 50 % with SPACE 1/16 turns 1/8 hats into 1/16s; above 50 % adds 1/32 ghosts.

— PROB decides, per incoming hit, whether its echoes play.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DENSITY M	0 % to 100 % default 0 %	0-50 %: one echo one SPACE later. 50-100 %: two more ghosts at half spacing.
SPACE	1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D default 1/16	Echo spacing: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D.
DARKEN	0 % to 100 % default 40 %	How much darker each generation of echoes is.
SWING	0 % to 100 % default 0 %	Delays the first echo.
PROB	0 % to 100 % default 100 %	Chance each incoming hit gets its echoes.
MIX	0 % to 100 % default 100 %	Echo level.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SHORTEN	0 % to 100 % default 30 %	Echoes carry only the start of each hit (tighter ghosts).

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DRY	ON / OFF default ON	Passes the input. Turn it off on a return track, so the return gives only the echoes. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DELAY

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Tape-style dub delay. Changing TIME bends the pitch as the tape head moves; feedback above 100 % saturates instead of exploding; THROW and FREEZE for live moves. Ducks its repeats around the kick.



USING IT

— As a return (DRY off): INPUT THROW ONLY and THROW mapped to a button gives dub throws.

— Turn TIME during a tail for the tape-bend. SLEW sets how long the bend takes.

— FEEDBACK past 100 % with TONE swept = the runaway dub effect, safe.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TIME	1 BAR, 1/2, 1/4, 1/4D, 1/4T, 1/8 ... (14) default 1/8D	Delay time, synced: 1 BAR to 1/32, then 5/16 and 7/16 (odd lengths: the repeats walk across the beat). Changing it slews the head (pitch bend).
FEEDBACK M	0 % to 115 % default 45 %	Repeats, up to 115 %. Above 100 % the loop saturates instead of growing.
TONE	LP 100 to HP 100 default OFF	Bipolar filter inside the feedback loop: left darker, right thinner. Each repeat goes further than the last.
DRIVE	0 % to 100 % default 20 %	Saturation inside the feedback loop (2x oversampled): each repeat dirtier than the last.
THROW	HOLD	Hold to send into the delay (louder send in INPUT ALWAYS, the only send in THROW ONLY).
FREEZE	ON / OFF default OFF	Loops the current repeats forever, losslessly: no feedback loss, no filtering, the tape wow stops (10 ms crossfade in and out); input muted. Turning TIME while frozen bends the loop (a tape-speed move) and it locks again when the head settles.
MIX M	0 % to 100 % default 30 %	Wet level.
WIDTH	0 % to 100 % default 50 %	Stereo width of the repeats.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SLEW	10 ms to 3.00 s default 400 ms	How long a TIME change takes (the length of the pitch bend).
MODE	STEREO, PING-PONG default PING-PONG	STEREO or PING-PONG.
INPUT	ALWAYS, THROW ONLY default ALWAYS	ALWAYS, or THROW ONLY (nothing enters except while THROW is held).
DUCK	0 % to 100 % default 0 %	Pocket on the repeats: they duck under the kick.
DUCK SHAPE	0 % to 100 % default 60 %	Pocket shape for the repeats.
WOW	0 % to 100 % default 10 %	Tape wow on the repeats.
LO CUT	20 Hz to 600 Hz default 80 Hz	Low cut inside the loop.

SETUP

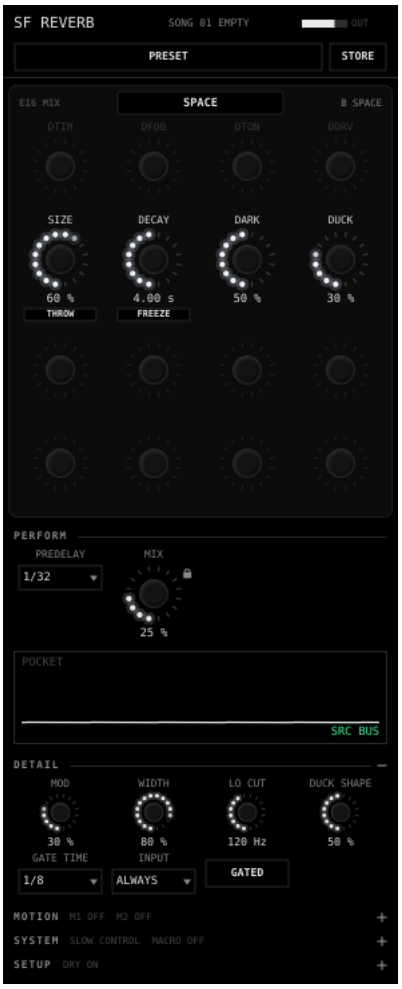
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DRY	ON / OFF default ON	Passes the input at unity (MIX sets only the repeats). Turn it off on a return track. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF REVERB

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Dark feedback-network reverb with freeze, throw, gated mode and ducking (kick pocket plus dry-follow, so tails bloom in the gaps).



USING IT

— DUCK at 30-60 % keeps big spaces from washing out the groove.

— FREEZE holds the space as a drone; THROW sends a single hit.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIZE M	0 % to 100 % default 60 %	Room size.
DECAY M	0.30 s to 30.00 s default 4.00 s	Decay time of the lows, 0.3 to 30 s (measured: DECAY is the real RT60).
DARK	0 % to 100 % default 50 %	Damping: shortens the decay of the highs only, so the tail gets darker without getting shorter.
PREDELAY	0, 1/64, 1/32, 1/16, 1/8, 1/4 default 1/32	Synced predelay: 0, 1/64, 1/32, 1/16, 1/8, 1/4.
DUCK	0 % to 100 % default 30 %	Ducks the tail under the kick and while the dry signal plays.
FREEZE	ON / OFF default OFF	Infinite hold: the space keeps its level and brightness (measured over 14 s). The input fades into the frozen space over 10 ms, then is muted.
THROW	HOLD	Hold to send (or the only send in INPUT THROW ONLY).
MIX M	0 % to 100 % default 25 %	Wet level.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
MOD	0 % to 100 % default 30 %	Modulation of the tail (chorus-like movement).
WIDTH	0 % to 100 % default 80 %	Stereo width.
GATED	ON / OFF default OFF	Gated reverb: the tail is cut GATE TIME after each hit.
GATE TIME	1/16, 1/8, 1/4, 1/2 default 1/8	1/16, 1/8, 1/4, 1/2.
LO CUT	20 Hz to 800 Hz default 120 Hz	Low cut on the input.
DUCK SHAPE	0 % to 100 % default 50 %	Pocket shape for the tail.
INPUT	ALWAYS, THROW ONLY default ALWAYS	ALWAYS or THROW ONLY.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DRY	ON / OFF default ON	Passes the input at unity (MIX sets only the tail). Turn it off on a return track. Not stored in songs or presets.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF TERROIR

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Convolution reverb for real places. Eight PLACE slots, each a space you captured (a clap, a balloon, or the sweep file played from a speaker and recorded; in the set, through SF CAPTURE on the room mics' track) or any IR file, kept inside the Live set and recalled with songs. Its first eight controls are SF REVERB's, so it swaps in on a return.



USING IT

— Tie a song to a place: choose its PLACE (the boxes on the display), then STORE the song. A new place crossfades in with the tail it already has, never cut off.

— On B SPACE instead of SF REVERB: the E16's SPACE page fits as it is. Map SIZE, DECAY, DARK, DUCK and the two presses to it once (Cmd-M).

— LENGTH 1/4 or 1/8 turns a place into a gated hit in time; REVERSE with LENGTH 1/2 gives a swell that lands on the next beat.

— Stereo from one microphone: record the same sweep or clap from two spots (or a shotgun aimed at two parts of the room), name the files ... L and ... R, and drop both at once. Each take lines up at its own onset; their tails differ, so the place is wide (WIDTH under 100 % if too wide).

— The room you film in, on the whole mix: capture it from the camera's place with the speaker where you play, then put SF TERROIR on the recording (DRY on; preset MIX/IN THE ROOM): the sound sits in the room the picture shows.

CAPTURING A PLACE

SOURCE	HOW
CLAP	Clap once (or burst a balloon) and record a few seconds of the room. Drop the recording: the loudest event is used, from 1 ms before its first arrival to where its tail meets the noise (and stays there for 300 ms), cut before a second clap as loud as the first; when that cut shortens the tail, the status line says where (CUT AT ... S: A SECOND CLAP). Quicker than the sweep, but the noise takes its tail sooner: a good backup.
SWEEP	SWEEP FILE (on the display) writes SF TERROIR SWEEP.wav (Music/SF Live Suite/TERROIR): 43 s, three 10 s sweeps, the lowest octaves softer so a small speaker isn't pushed below what it can play (the analysis evens them out again). Play it from any speaker (a phone into a MiniRig is fine; by cable rather than Bluetooth if you can), record all of it (32-bit float, or under full scale: a clipped take blurs the place), drop the recording. The sweeps are found, the speaker's clock drift measured and corrected, the three averaged: the cleanest IR, and the speaker's distortion left out. A recording of one sweep alone through a small speaker can pass for a clap: set IMPORT to SWEEP.
IN THE SET	SF CAPTURE on the room mics' track (ROOM in SF SHOW) plays the sweep file out of its track's output, a speaker on it, and records the mics on the same clock. Transport stopped: click CAPTURE (top right of the display, amber), then the slot to fill. 46 s, counted on the status line, then the take is analysed as the sweep, never as anything else (if it didn't come back: THE SWEEP DIDN'T COME BACK, and the slot stays as it was); the slot takes the name ROOM and the date and time, and the take stays in Music/SF Live Suite/TERROIR/Captures. CAPTURE again (it reads STOP) ends it early, and so do the transport starting, PANIC and Live's audio restarting, whatever the take had reached (still waiting for SF CAPTURE, playing, or being written): nothing is stored and the slot stays as it was. Setting up: SF CAPTURE.
IR FILE	Any impulse response (WAV, AIFF, FLAC), mono or stereo, any rate, from a few ms to 12 s (longer is cut). It comes in as it is: from 1 ms before its first arrival (however long before the strongest, a click at the very first sample too) to its last moment over the floor, every arrival however strong and every gap kept, every channel's timing and polarity too; only its last 10 ms fade out. With AUTO, a file with nothing before its first arrival (silence, or under 5 ms) is taken for an IR file; IMPORT IR FILE says so for any file.
THE RESULT	Every place is scaled to the same energy (near SF REVERB's level, the same at any rate it was recorded or is played at), high-passed at 20 Hz (0.1 dB at 100 Hz on a very short IR) and stored in the set as FLAC. A file with nothing left (or samples that are not numbers) is refused. The status line gives its kind (SWEEP x3, IR FILE or CLAP: what IMPORT AUTO took the file for), length, RT60 and how far it rises over the noise (under 30 dB: amber, try again closer or louder). A stereo take keeps the time and level difference between its mics. From a recorder's poly file the mics are used: tracks named MIX (a MixPre's MixL and MixR) are skipped, then the first two. Two channels carrying the same signal make a mono place.

IN THE FIELD

STEP	WHAT
MICS	Two mics recorded at once keep the room's own width. Cardioids in ORTF (capsules 17 cm apart, 110° between them) sound natural on headphones; 40 cm to 1 m apart, bigger. Pointed away from the speaker they hear more room and less speaker. Omnis: 40 cm to 1 m apart. A shotgun (a 416) gives an aimed, coloured place; one mic makes stereo in two takes (USING IT). Pads at 0, low cut off.
SPEAKER	Where the sound should come from, aimed at a wall or into the space, standing where nothing buzzes. A small one is enough: the lows it can't play are lows a techno reverb cuts anyway. A bigger one plays louder, so a long tail stays over the noise longer (10 dB more: a sixth of the decay). A little distortion stays out of the place (the sweep sets it apart); a rattle doesn't. Avoid a speaker that adapts its sound as it plays; bass boost off.
PLAYER	Play the sweep file from the phone's file app, not a music app (no normalising, spatial audio or EQ), in airplane mode. By cable if you can; Bluetooth works (its clock is corrected), but a dropout spoils a sweep. As loud as the speaker goes without buzzing.
RECORDER	48 kHz, 32-bit float. The same gain on both mics, low cut and limiters off. Input 1 is the mic on your left as you face the speaker. The mix and its pan don't matter: SF TERROIR reads the mic tracks.
THE TAKE	Start recording, then press play. Keep still and quiet for the 43 s; stop a few seconds after the file ends. Take a second spot deeper in the room: there are eight slots. A minute of the place's own sound makes an ambience bed for a film.
BACK HOME	Name the file after the place (the slot takes its name). Choose a PLACE slot, then LOAD (top right of the plugin; for an L and an R take, select both). Dropping works too, but Live hides plugin windows while the Finder is in front: drag from Live's browser instead. The status line should read SWEEP x3 (from a poly file, FROM and the two mic tracks) and 30 dB or more over the noise. A sweep take that comes back as CLAP or IR FILE wasn't found as a sweep: play it louder or closer (a take of one sweep: IMPORT SWEEP).
IN THE MIX	START at 5-20 ms drops the speaker's direct sound and leaves the room; it also evens out a place that leans to one side because the speaker stood nearer one mic. WIDTH under 100 % if two takes from one mic come out too wide; DARK if the mics or the speaker made the place bright.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIZE M	50 % to 200 % default 100 %	The same place bigger (slower and lower) or smaller: the IR resampled, 50-200 %. While it moves, a new IR is built at most every 40 ms and crossfaded in; each one asked for later than the last that went in, so the place follows the knob and ends where it stands (after a render, a new sample rate or a reload, the knobs' own IR is in before the first note).
DECAY M	10 % to 100 % default 100 %	Shortens the tail: the RT60 times DECAY. 100 % = as captured.
DARK	0 % to 100 % default 0 %	The tail darkens as it goes, up to 8 octaves a second at 100 %. 0 = as captured.
PREDELAY	0, 1/64, 1/32, 1/16, 1/8, 1/4 default 0	Synced predelay: 0, 1/64, 1/32, 1/16, 1/8, 1/4.
DUCK	0 % to 100 % default 30 %	Ducks the tail under the kick and while the dry signal plays.
FREEZE	ON / OFF default OFF	Holds the wet sound's spectrum as an endless texture (in over 150 ms, out over 400 ms) while the tail dies away; the input is muted while it holds.
THROW	HOLD	Hold to send (or the only send in INPUT THROW ONLY).
MIX M	0 % to 100 % default 25 %	Wet level.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PLACE	1, 2, 3, 4, 5, 6, 7, 8 default 1	Slot 1-8 (or click the boxes on the display). An empty slot plays the built-in room. Stored in songs; presets never change it.
LENGTH	FULL, 4 BAR, 2 BAR, 1 BAR, 1/2, 1/4, 1/8, 1/16 default FULL	Cuts the IR to a note value (FULL, 4 BAR to 1/16), with a raised-cosine end, at Live's tempo and meter from the first note (also with a preset saved at another tempo).
REVERSE	ON / OFF default OFF	Plays the IR backwards: a swell that ends on the hit, LENGTH long.
START	0.0 ms to 250 ms default 0.0 ms	Skips the start of the IR, 0-250 ms: the direct sound and early reflections go, the tail stays.
WIDTH	0 % to 200 % default 100 %	Stereo width of the wet signal, 0-200 %. A mono capture is spread by a decorrelated copy of its tail.
LO CUT	20 Hz to 800 Hz default 120 Hz	Low cut on the input.
DUCK SHAPE	0 % to 100 % default 50 %	Pocket shape for the tail.
INPUT	ALWAYS, THROW ONLY default ALWAYS	ALWAYS or THROW ONLY.

SETUP

CONTROL	RANGE / DEFAULT	WHAT IT DOES
DRY	ON / OFF default ON	Passes the input at unity (MIX sets only the tail). Turn it off on a return track. Not stored in songs or presets.
IMPORT	AUTO, IR FILE, CLAP, SWEEP default AUTO	What a dropped or loaded file is. AUTO decides, and the status line says what it took it for; IR FILE keeps a finished impulse response as it is (every arrival, however strong); CLAP cuts a room recording before a second clap and says where; SWEEP takes a recording of the sweep file (one sweep through a small speaker, which AUTO can take for a clap). SF CAPTURE's takes are always the sweep. How files come in, not a sound: in no song or preset.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF CAPTURE

AUDIO EFFECT

The room mics' track, for capturing a place in the set. It never passes the mics: its output is silence, or the sweep while SF TERROIR captures. A take plays the sweep file out of this track's output (a speaker on it) and records the mics on the same clock; SF TERROIR analyses it into the PLACE slot it asked for.



USING IT

— On an audio track with the room mics as its input, a stereo pair (ROOM in SF SHOW: Ext. In 1/2), Monitor In: a track's devices hear its input only while it monitors. Its output goes where the speaker plugs in (ROOM: 3/4, the headphone out), never to Main; its sends disabled.

— One in the set. Any SF TERROIR in the set asks it for a take (on B SPACE in place of SF REVERB, or on a return of its own): CAPTURE on TERROIR's display, then a PLACE slot.

— Live's transport must be stopped: a take won't start while it plays, and stops (a 10 ms fade, nothing stored) if it starts, Live renders or PANIC is pressed. PANIC, or Live's audio restarting (a new sample rate or buffer), ends a take asked for before it whatever it had reached, even one still waiting to start or being written: no sweep after it, no file kept. Capture at soundcheck, or after the set.

— IN L and IN R show the mics. To record the room with the set, arm the track with the others: Live records the mics as they come in, before SF CAPTURE, so the room is in the Arrangement for the mix.

— Every take stays in Music/SF Live Suite/TERROIR/Captures, named ROOM and the date and time it was asked for (32-bit float WAV): drop it on another slot, or in another set, any time.

IN THE FIELD

STEP	WHAT
MICS	The pair on inputs 1 and 2, input 1 the mic on your left as you face the speaker; in Live's Settings > Audio > Input Config, 1/2 as a stereo pair. The same gain on both, low cut off. Placed as SF TERROIR's IN THE FIELD, MICS.
SPEAKER	On the outputs the track goes to (SF SHOW: the headphone out; a MiniRig's line in on it is fine), where the sound should come from, aimed at a wall or into the room. The mics must not reach that output in the interface's own mixer (TotalMix), or the speaker plays the mics and feeds back.
LEVEL	As loud as the speaker goes without buzzing: turn the speaker and the headphone level, not SWEEP LEVEL, which is for headroom. The sweep goes through the track's fader and on/off too (SF SHOW: 0 dB, on). IN L and IN R well under the top: a take whose mics clip says so on both plugins; turn down and take it again.
THE TAKE	Transport stopped. In SF TERROIR click CAPTURE (amber), then the slot to fill: 46 s, the sweep file and 3 s after it. Keep the room quiet; TERROIR's status line counts the seconds. CAPTURE again (it reads STOP) ends it early, while it plays or is being written: the sweep fades out in 10 ms and nothing is stored.
THE RESULT	SF TERROIR analyses the take as the sweep: the slot takes the name ROOM and the date and time, and the status line should read SWEEP x3 and 30 dB or more over the noise. Under that: the speaker louder or the mics closer, and again. THE SWEEP DIDN'T COME BACK: the speaker, the mics or the routing (the slot stays as it was). NO SWEEP IN THE RIGHT (or LEFT) CHANNEL: that mic heard nothing, and the place is mono from the other.
REFUSED	TERROIR's status line says why: the transport is playing; NO SF CAPTURE (none in the set, or its device off); a capture already running; SF CAPTURE DIDN'T ANSWER (its device off, or Live's audio stopped). A take ends with CAPTURE STOPPED when SF CAPTURE stops answering (removed, or its track's audio stood still: STOPPED: THIS TRACK'S AUDIO STOOD STILL on its display) or the transport starts; nothing is stored, and the next take can be asked for at once.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SWEEP LEVEL	-30.0 dB to 0.0 dB default -6.0 dB	The sweep's peak at the output, -30 to 0 dBFS (-6 is the file's own). For headroom only: loudness comes from the speaker. Not stored in songs or presets, never glides.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DRIVE

AUDIO EFFECT LATENCY 5 SAMPLES AT 48 KHZ

Saturation with one CHARACTER knob from smooth to fuzz: TAPE, TUBE, FUZZ, INDUSTRIAL. 4x oversampled, auto gain.



USING IT

— Your smooth-then-heavy arc on a single knob: automate CHARACTER across a track.

— LOW KEEP keeps the sub clean while the rest distorts.

PERFORM

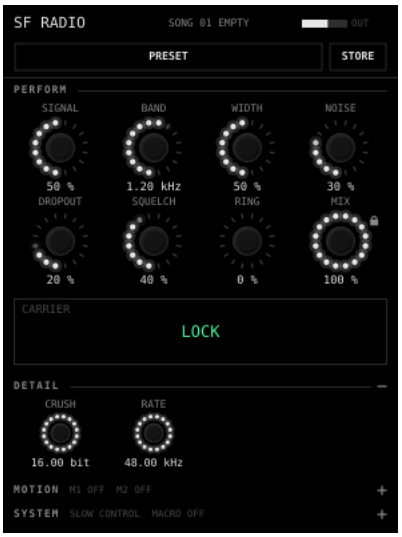
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DRIVE M	0 % to 100 % default 20 %	Amount, 0 to +36 dB into the shaper.
CHARACTER M	0 % to 100 % default 0 %	TAPE > TUBE > FUZZ > INDUSTRIAL (wavefold).
TONE	-100 % to 100 % default 0 %	Tilt EQ around 800 Hz.
LOW KEEP	0 Hz to 300 Hz default 0 Hz	Below this frequency the signal stays clean. 0 = off.
MIX	0 % to 100 % default 100 %	Dry/wet. The dry signal runs through an identical oversampler, so the blend is phase-aligned (no comb filtering) and MIX 0 is the clean input.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.
AUTO GAIN	ON / OFF default ON	Holds the driven signal's loudness near the input's, before MIX.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF RADIO

AUDIO EFFECT

Transmission processor: band-limit, dirt, noise, dropouts and squelch. SIGNAL is one knob from lost to clear: it also takes CRUSH and RATE from their setting at 0 to clean at 100 %.



USING IT

— On a voice, ride SIGNAL from 0 to 100 % to bring a ghost transmission into focus.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
SIGNAL M	0 % to 100 % default 50 %	Clarity: band width, noise, dropouts, distortion, CRUSH and RATE all follow it.
BAND	300 Hz to 3.00 kHz default 1.20 kHz	Centre of the pass band.
WIDTH	0 % to 100 % default 50 %	Width of the pass band.
NOISE	0 % to 100 % default 30 %	Noise floor.
DROPOUT	0 % to 100 % default 20 %	How often the signal drops out (more at low SIGNAL). Each drop fades out and back in over about 10 ms.
SQUELCH	0 % to 100 % default 40 %	A squeal as each dropout ends: a whistle sliding down from the top of the band, a little breath around it, 0.1-0.2 s (longer with more SQUELCH). Fades in and out smoothly: no click.
RING	0 % to 100 % default 0 %	Ring modulation (47 Hz) for a metallic voice.
MIX	0 % to 100 % default 100 %	Dry/wet.

DETAIL

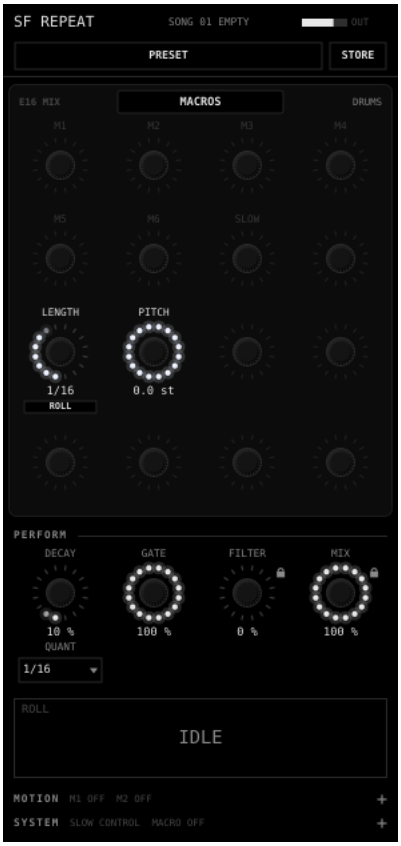
CONTROL	RANGE / DEFAULT	WHAT IT DOES
CRUSH	4.00 bit to 16.00 bit default 16.00 bit	Bit depth at SIGNAL 0; SIGNAL brings it back to 16 bits (16 = off).
RATE	2.00 kHz to 48.00 kHz default 48.00 kHz	Sample rate at SIGNAL 0; SIGNAL brings it back to full rate (48 kHz = off).

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF REPEAT

AUDIO EFFECT

Beat repeat for fills and builds. Hold ROLL: it captures LENGTH of audio from the next QUANT point and loops it, each pass quieter, lower, darker.



USING IT

— Map ROLL to a button; LENGTH 1/16 then 1/32 into a drop.

— PITCH negative = the classic falling roll.

PERFORM

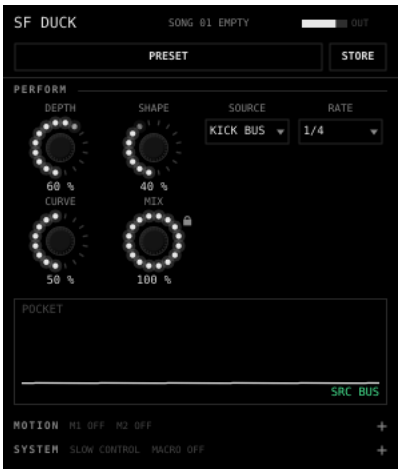
CONTROL	RANGE / DEFAULT	WHAT IT DOES
ROLL	HOLD	Hold to roll. The slice is captured from the QUANT point (up to 4 s) and kept for the whole hold. Release: the loop keeps playing while it crossfades back to the input.
LENGTH	1/4, 1/8, 1/16, 1/32, 1/8T, 1/16T default 1/16	Loop length: 1/4 to 1/32 and triplets.
DECAY	0 % to 100 % default 10 %	Level loss per pass.
PITCH	-12.0 st to 0.0 st default 0.0 st	Pitch drop over the repeats: falls by this amount every four passes.
GATE	0 % to 100 % default 100 %	How much of each pass plays.
FILTER	0 % to 100 % default 0 %	Darkens over the repeats.
MIX	0 % to 100 % default 100 %	Roll level over the input.
QUANT	OFF, 1/16, 1/8, 1/4 default 1/16	Start point: OFF (at once), 1/16, 1/8, 1/4, on the groove's grid: where SF KICK's kicks land, a little after Live's beat with a sequencer clocked by Live (Live's grid with no SF KICK).

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF DUCK

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Pocket as an effect for any audio: follows SF KICK on the bus, a sidechain input, or a tempo-synced volume shape.



USING IT

— Put it on any track that should pump with the kick: no sidechain routing needed.

— SOURCE SIDECHAIN for an external kick; GRID for a classic volume-shaper pump.

PERFORM

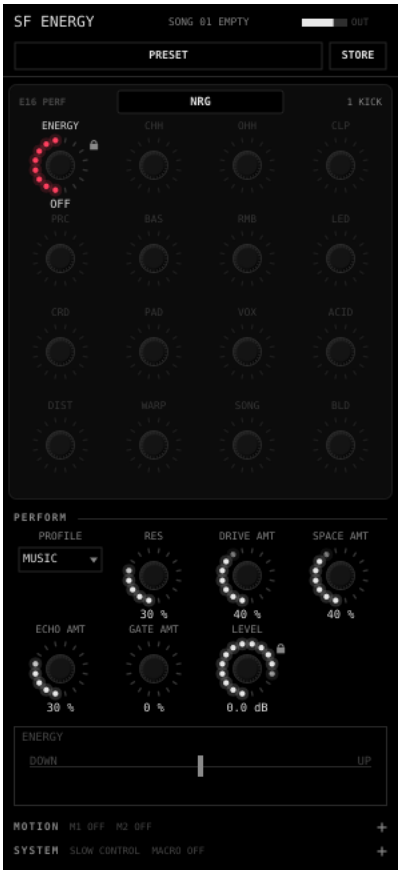
CONTROL	RANGE / DEFAULT	WHAT IT DOES
DEPTH M	0 % to 100 % default 60 %	How far it ducks.
SHAPE M	0 % to 100 % default 40 %	Transparent (pitch-tracking) to full pump.
SOURCE	KICK BUS, SIDECHAIN, GRID default KICK BUS	KICK BUS (SF KICK), SIDECHAIN (Live's sidechain input), GRID (volume shape).
RATE	1/2, 1/4, 1/8 default 1/4	GRID: 1/2, 1/4, 1/8, on the groove's beat (where SF KICK's kicks land; Live's beat with no SF KICK).
CURVE	0 % to 100 % default 50 %	GRID: shape of the recovery.
MIX	0 % to 100 % default 100 %	Amount.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF ENERGY

AUDIO EFFECT

Group energy. Replaces the DJ filter on groups: one bipolar knob, centre neutral, that adds energy as well as taking it away.



USING IT

— One per group (drums, bass, music, voice) with PROFILE to match; in the show template one on every instrument track too (the E16's NRG page).

— Down: low-pass, space and softness. Up: high-pass with a resonant edge, drive, 1/16 echo and chop.

— PROFILE DRUMS at zero takes the part out of the mix: a low-pass alone leaves a kick's thump, so KCK NRG all the way down is no kick.

— With SLOW on, an 8-bar high-pass build happens by itself.

REFERENCE

NAME	SOUND
PROFILE	DRUMS (high-pass side thins the kick moderately; at the very bottom the part is gone: the last eighth of the knob fades it out, space and echo with it), BASS (strong low-pass side, gentle high-pass), MUSIC (both sides full), VOICE, MASTER (gentle both ways).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ENERGY M	LP 100 to HP 100 default OFF	Bipolar. Centre = neutral. Left = down, right = up.
PROFILE	DRUMS, BASS, MUSIC, VOICE, MASTER default MUSIC	How far each side travels for this kind of group.
RES	0 % to 100 % default 30 %	Resonant edge (stronger on the up side).
DRIVE AMT	0 % to 100 % default 40 %	Drive added on the up side (2x oversampled, from clean at centre).
SPACE AMT	0 % to 100 % default 40 %	Reverb added on the down side. It rings out when the knob comes back past centre.
ECHO AMT	0 % to 100 % default 30 %	1/16 echo added on the up side.
GATE AMT	0 % to 100 % default 0 %	1/16 chop added on the up side, on the groove's sixteenths (where SF KICK's kicks land).
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

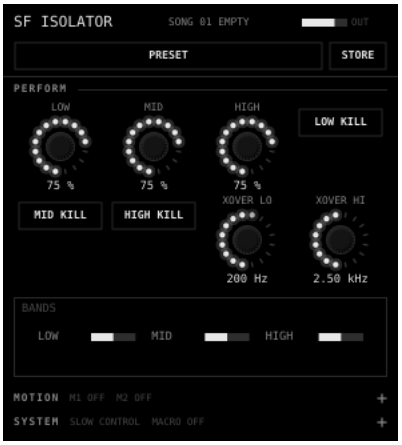
M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

10 MASTER AND MIX

SF ISOLATOR

AUDIO EFFECT

3-band DJ kill EQ (Linkwitz-Riley, phase-aligned).



USING IT

— Kill the lows before a drop, bring them back on the one.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LOW	0 % to 100 % default 75 %	0 = kill, 75 % = 0 dB, 100 % = +6 dB.
MID	0 % to 100 % default 75 %	Same for mids.
HIGH	0 % to 100 % default 75 %	Same for highs.
LOW KILL	ON / OFF default OFF	Kill the low band.
MID KILL	ON / OFF default OFF	Kill the mid band.
HIGH KILL	ON / OFF default OFF	Kill the high band.
XOVER LO	60 Hz to 500 Hz default 200 Hz	Low/mid crossover.
XOVER HI	1.00 kHz to 8.00 kHz default 2.50 kHz	Mid/high crossover.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF BUILD

AUDIO EFFECT

One knob for a whole-mix build: high-pass sweep, noise riser, reverb wash and width. DROP is played: press it and the music stops at its source, leaving the reverb and echo tails; press it again and the drop comes in on the next kick.



USING IT

— Turn BUILD up over the build. Press DROP: from the next sixteenth every SF instrument stops at its source and the returns ring out: SPACE's reverb, DUB's echoes, and BUILD's own wash and noise for TAIL (the build ends there). The display shows CUT. The gap is as long as you play it.

— Press DROP again: the drop comes in on the next beat after the cut, on the kick (DROP ON KICK, then RETURN KNOB). The parts come back 10 ms before the kick, so it keeps its attack, and a part with a note up to 30 ms early comes back just before that note. A third press while the drop waits changes nothing; a press after the drop cuts again. Holding DROP is one press.

— What stops: every SF instrument (KICK, PERC, STRIKE, RUMBLE, BASS, LEAD, PAD, VOICE), and SF STRIP and TAP unless their ROLE is DECK. What plays on: SF DECK, the returns, and tracks without an SF instrument (give them SF STRIP or TAP to stop with the rest). The parts keep playing silently, so the drop comes in mid-pattern, as a mute would. SF KICK reports no kick while silent, so nothing ducks to it: the tails ring evenly.

— LAND ON KICK (the default): the cut and the drop follow where SF KICK's kicks land against Live's beat (a sequencer clocked by Live plays late: the OXI One about 40 ms), and BUILD's readout under its bar shows where the kick reaches it (KICK +80 MS). Until the kick is found, and on BAR LINE, Live's beat.

— The cut ends at once when Live stops, when the transport jumps back (an Arrangement loop wraps), on PANIC, and within a second if BUILD is switched off or deleted.

— TAIL: the riser's noise and the wash ring on for that long after the build ends at the cut, fading out; longer than the gap, it carries into the drop, DROP DUCK lower. CUT: gone in 5 ms.

— DROP DUCK (off by default): for DUCK TIME after the drop, every SF part with Pocket dips this far under each kick, whatever its POCKET DEPTH, so the first kicks land on top of everything that comes back. It takes the bass, rumble and pads from around those kicks: the kick stands alone, with less weight than a kick brought back by hand. Full depth for the first half of DUCK TIME, easing out over the second. The parts: BASS, LEAD, PAD, VOICE, RUMBLE, the wet signal of DELAY and REVERB, DUCK, and TAP and STRIP unless their ROLE is KICK, PERC or DECK. It needs SF KICK reporting (KICK REPORTING on SF CONTROL). 0 dB: off.

— After a drop the build stays off until BUILD is back below 5 % (display: RETURN KNOB), so you can bring the knob down without re-building. Looping back over a drop doesn't duck again. UNUSED in Live's parameter list was GAP (the gap is played now).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
BUILD M	0 % to 100 % default 0 %	The build amount.
DROP	HOLD	First press: the cut (every SF instrument stops from the next sixteenth; the tails ring). Second press: the drop, on the next kick.
NOISE	0 % to 100 % default 40 %	Noise riser level.
SPACE	0 % to 100 % default 40 %	Reverb wash.
WIDTH	0 % to 100 % default 40 %	Stereo widening.
HP MAX	200 Hz to 4.00 kHz default 1.20 kHz	How high the high-pass goes at full build.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.
TAIL	CUT, 1/16, 1/8, 1/4, 1/2, 1 BAR, 2 BARS default 1/8	How long the noise and the wash ring on after the build ends at the cut, fading out: CUT (5 ms), 1/16 to 2 BARS.

DETAIL

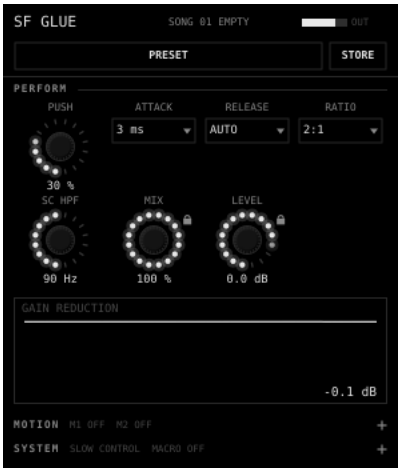
CONTROL	RANGE / DEFAULT	WHAT IT DOES
UNUSED	OFF, 1/16, 1/8, 1/4, 1/2, 1 BAR default OFF	Does nothing (it was GAP: the gap is played with DROP now). Kept so older sets load.
DROP DUCK	0.0 dB to 12.0 dB default 0.0 dB	How far every SF part with Pocket dips under each kick after the drop, and how much lower the tail runs into it. 0 = off (the default).
DUCK TIME	1/4, 1/2, 1 BAR, 2 BARS, 4 BARS default 1 BAR	How long the drop duck lasts after the drop: full depth for the first half, easing out over the second.
LAND ON	KICK, BAR LINE default KICK	KICK (the default): the cut and the drop follow where SF KICK's kicks land against Live's beat; the drop comes in 10 ms before the kick. BAR LINE: Live's beat, whatever the kick does.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF GLUE

AUDIO EFFECT

Bus compressor with one PUSH knob (threshold and makeup together). 6 dB soft knee.



USING IT

— PUSH 20-40 % for glue; more for audible pump.

PERFORM

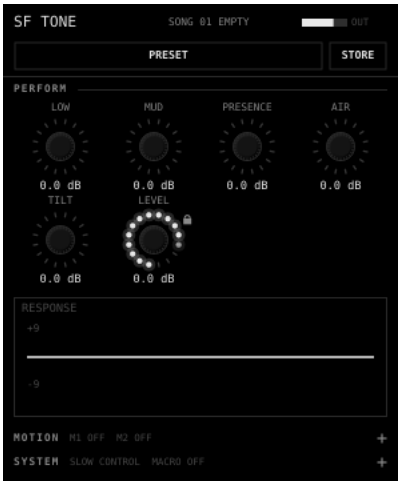
CONTROL	RANGE / DEFAULT	WHAT IT DOES
PUSH M	0 % to 100 % default 30 %	Compression and makeup together.
ATTACK	0.1 ms, 0.3 ms, 1 ms, 3 ms, 10 ms, 30 ms default 3 ms	0.1 to 30 ms.
RELEASE	0.1 s, 0.3 s, 0.6 s, 1.2 s, AUTO default AUTO	0.1 to 1.2 s or AUTO (dual-stage).
RATIO	2:1, 4:1, 10:1 default 2:1	2:1, 4:1, 10:1.
SC HPF	20 Hz to 300 Hz default 90 Hz	High-pass on the detector so the kick doesn't drive it.
MIX	0 % to 100 % default 100 %	Parallel compression.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF TONE

AUDIO EFFECT

Master voicing EQ: five broad moves on one row (low shelf, mud cut, presence, air, tilt). The ranges stop before damage. Runs at 2x, so the air shelf keeps its shape up to 20 kHz.



USING IT

— After GLUE, before LOWEND and LIMIT.

— MUD 1-3 dB cleans a dense mix without thinning it. LOW and AIR at +1-2 dB open up a big system. TILT darkens or brightens everything in one move.

— For the room or PA itself use SF ROOM EQ (stored per venue, after the recording point); TONE is the sound of the set.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
LOW	-6.0 dB to 6.0 dB default 0.0 dB	Low shelf at 70 Hz, +/-6 dB.
MUD	0.0 dB to 6.0 dB default 0.0 dB	Broad cut at 280 Hz, 0 to 6 dB.
PRESENCE	-4.0 dB to 4.0 dB default 0.0 dB	Bell at 3.2 kHz, +/-4 dB.
AIR	-6.0 dB to 6.0 dB default 0.0 dB	High shelf at 12 kHz, +/-6 dB.
TILT	-3.0 dB to 3.0 dB default 0.0 dB	Tilts the whole spectrum around 900 Hz: right brighter, left darker (+/-3 dB at the ends).
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF LOWEND

AUDIO EFFECT

Low-end safety: mono below a frequency, sub clip, rumble cut, width above.



USING IT

— Last-but-one on the master, before LIMIT.

PERFORM

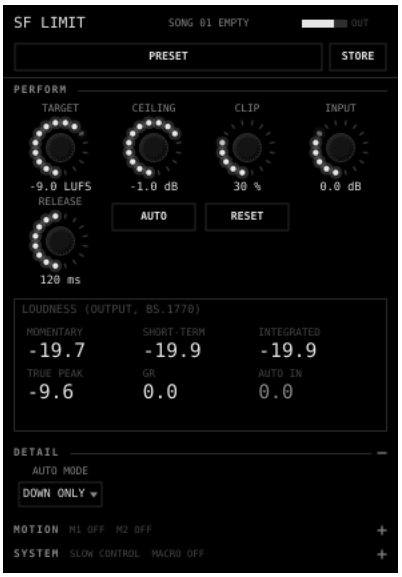
CONTROL	RANGE / DEFAULT	WHAT IT DOES
MONO FREQ	40 Hz to 300 Hz default 120 Hz	Everything below this is mono.
SUB CLIP	0 % to 100 % default 20 %	Soft clip on the mono low band (louder sub, same peak), 2x oversampled.
RUMBLE CUT	10 Hz to 45 Hz default 25 Hz	24 dB/oct high-pass for DC and rumble.
WIDTH	0.00 x to 2.00 x default 1.00 x	Stereo width above MONO FREQ.
LOW TRIM	-12.0 dB to 6.0 dB default 0.0 dB	Level of the low band.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF LIMIT

AUDIO EFFECT LATENCY 176 SAMPLES AT 48 KHZ

Clipper and true-peak lookahead limiter with a BS.1770 loudness meter. The clipper runs at 4x through linear-phase oversampling; the limiter detects true peaks at 8x and never lets the output past CEILING. AUTO trims the input so short-term loudness sits at TARGET.



USING IT

— Last on the master. CEILING -1 dB for venues and releases.

— AUTO keeps the loudest moments of the last minute at TARGET, moving the input by at most 1 dB per second, so a whole set sits at TARGET without anyone riding the master. A breakdown doesn't count as the set getting quieter: AUTO holds through it, and the drop comes back where it was. AUTO MODE DOWN ONLY (the default) never lifts above unity; UP AND DOWN also lifts a quieter set (0.5 dB per second), but never a breakdown.

— CLIP takes the first few dB off the peaks transparently (kick transients) so the limiter works less.

— Recording a set: RESET at the start; INTEGRATED then reads the loudness of the whole set.

— Latency 176 samples at 48 kHz, compensated by Live.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
TARGET	-16.0 LUFS to -5.0 LUFS default -9.0 LUFS	Loudness AUTO aims for (LUFS short-term).
CEILING	-3.0 dB to -0.1 dB default -1.0 dB	True-peak ceiling (dBTP). The output never exceeds it, including between samples.
CLIP	0 % to 100 % default 30 %	Soft clip before the limiter, 4x oversampled, linear phase. The knee starts up to 6 dB below CEILING.
INPUT	-12.0 dB to 24.0 dB default 0.0 dB	Input gain.
RELEASE	20 ms to 500 ms default 120 ms	Limiter release. Under sustained limiting a slower envelope holds the gain, so dense passages don't pump.
AUTO	ON / OFF default OFF	Automatic input trim (+/-12 dB) that keeps the loudest short-term loudness of the last minute at TARGET. It moves up only after 3 s of music within 6 dB of TARGET, so it never lifts a breakdown. Measured on the raw input, so it settles without overshoot.
RESET	HOLD	Resets the integrated loudness and true-peak hold.

DETAIL

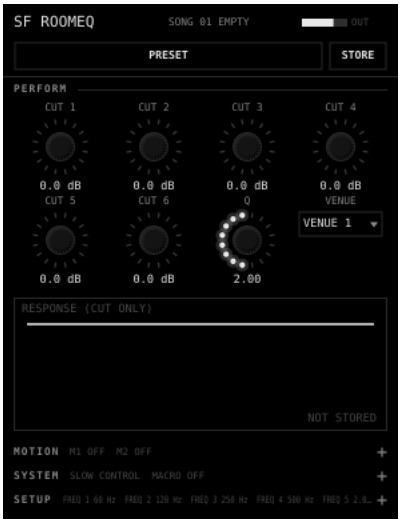
CONTROL	RANGE / DEFAULT	WHAT IT DOES
AUTO MODE	DOWN ONLY, UP AND DOWN default DOWN ONLY	DOWN ONLY (default): AUTO never lifts above unity, it only stops the set running above TARGET. UP AND DOWN: it also lifts a quieter set.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF ROOM EQ

AUDIO EFFECT

Cut-only correction for the room or PA, stored per venue (16 venues). Not part of songs.



USING IT

— Put it after the recording point: on an audio track PA with Audio From = Main, Post Mixer (the Main output), routed to the PA outputs, with a safety SF LIMIT after it (see Setting up Live). The set is recorded on its own track from the same signal, without the venue's correction.

— Find the boomy and harsh frequencies at soundcheck, cut them, STORE VENUE.

— Next time at that venue, choose it from VENUE.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
CUT 1	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 1 (0 to 18 dB).
CUT 2	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 2 (0 to 18 dB).
CUT 3	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 3 (0 to 18 dB).
CUT 4	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 4 (0 to 18 dB).
CUT 5	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 5 (0 to 18 dB).
CUT 6	0.0 dB to 18.0 dB default 0.0 dB	Cut at band 6 (0 to 18 dB).
Q	0.50 to 8.00 default 2.00	Width of all six bands.
VENUE	VENUE 1, VENUE 2, VENUE 3, VENUE 4, VENUE 5, VENUE 6 ... (16) default VENUE 1	Recalls a stored venue.

SETUP

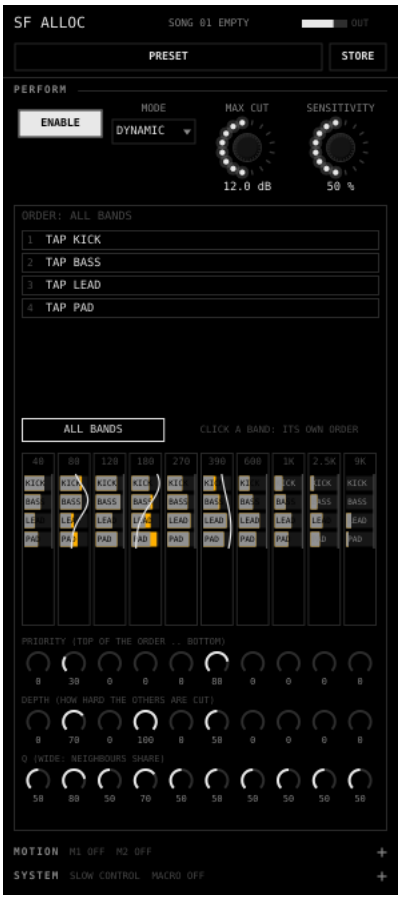
CONTROL	RANGE / DEFAULT	WHAT IT DOES
FREQ 1	30 Hz to 16.00 kHz default 60 Hz	Frequency of band 1.
FREQ 2	30 Hz to 16.00 kHz default 120 Hz	Frequency of band 2.
FREQ 3	30 Hz to 16.00 kHz default 250 Hz	Frequency of band 3.
FREQ 4	30 Hz to 16.00 kHz default 500 Hz	Frequency of band 4.
FREQ 5	30 Hz to 16.00 kHz default 2.00 kHz	Frequency of band 5.
FREQ 6	30 Hz to 16.00 kHz default 6.00 kHz	Frequency of band 6.
STORE VENUE	HOLD	Stores the current bands into the selected venue.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF ALLOC

AUDIO EFFECT

The mix assigner. You put the sources in order; each of the ten bands has a priority curve over that order, set with three knobs, and the display shows exactly what each source keeps and loses in that band. Pass-through audio: put it on the master.



USING IT

— Every SF instrument, SF STRIP, SF TAP and SF DECK appears in the SOURCES list by its Live track name (VST3) or its role (PERC, PERC 2 ...).

— Drag the list into order, top = most important. This is the ORDER: ALL BANDS. It is saved with the set and never changes on its own; new sources are added at the bottom. The order follows each plugin, not its name: reopening the set or renaming a track keeps every source's place, and a duplicated track joins at the bottom.

— Any band can have its own order: click the band (its frame turns white, the list shows ORDER: 180 HZ), drag, and the band gets a dot: its own order. LINK TO ALL puts it back on the global order. Click the band again or ALL BANDS to return to the global list.

— In each band: PRIORITY moves the curve's peak up and down the list, DEPTH sets how hard everything away from the peak is cut (0 = nothing), Q sets the width (narrow: one source owns the band; wide: neighbours share).

— Each band shows its sources in its own order, named in the lanes: grey = what plays, amber = what ALLOC takes away; the curve shows where the peak sits.

— MODE DYNAMIC cuts a source only while a more favoured source is actually playing in that band, so when the bass stops the pad gets 180 Hz back within a fraction of a second. STATIC cuts all the time, like a fixed EQ.

— The window, top to bottom: the order, the ten bands side by side, then a row each of PRIORITY, DEPTH and Q, one small knob per band (drag up and down). The 30 band knobs are not meant for hardware, but all are automatable in Live.

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ENABLE	ON / OFF default ON	Off = no cuts anywhere.
MODE	DYNAMIC, STATIC default DYNAMIC	DYNAMIC (cut only while a more favoured source plays in the band) or STATIC (always).
MAX CUT	0.0 dB to 24.0 dB default 12.0 dB	The largest cut any source can get (DEPTH 100 %, far from the peak).
SENSITIVITY	0 % to 100 % default 50 %	DYNAMIC: how quiet a favoured source can be and still count as playing.

DETAIL

CONTROL	RANGE / DEFAULT	WHAT IT DOES
PRIORITY 1	0 % to 100 % default 0 %	Band 1: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 2	0 % to 100 % default 0 %	Band 2: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 3	0 % to 100 % default 0 %	Band 3: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 4	0 % to 100 % default 0 %	Band 4: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 5	0 % to 100 % default 0 %	Band 5: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 6	0 % to 100 % default 0 %	Band 6: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 7	0 % to 100 % default 0 %	Band 7: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 8	0 % to 100 % default 0 %	Band 8: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 9	0 % to 100 % default 0 %	Band 9: where the priority peak sits in the list (0 = top, 100 = bottom).
PRIORITY 10	0 % to 100 % default 0 %	Band 10: where the priority peak sits in the list (0 = top, 100 = bottom).
DEPTH 1	0 % to 100 % default 0 %	Band 1: how hard sources away from the peak are cut.
DEPTH 2	0 % to 100 % default 0 %	Band 2: how hard sources away from the peak are cut.
DEPTH 3	0 % to 100 % default 0 %	Band 3: how hard sources away from the peak are cut.
DEPTH 4	0 % to 100 % default 0 %	Band 4: how hard sources away from the peak are cut.
DEPTH 5	0 % to 100 % default 0 %	Band 5: how hard sources away from the peak are cut.
DEPTH 6	0 % to 100 % default 0 %	Band 6: how hard sources away from the peak are cut.
DEPTH 7	0 % to 100 % default 0 %	Band 7: how hard sources away from the peak are cut.
DEPTH 8	0 % to 100 % default 0 %	Band 8: how hard sources away from the peak are cut.
DEPTH 9	0 % to 100 % default 0 %	Band 9: how hard sources away from the peak are cut.
DEPTH 10	0 % to 100 % default 0 %	Band 10: how hard sources away from the peak are cut.
Q 1	0 % to 100 % default 50 %	Band 1: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 2	0 % to 100 % default 50 %	Band 2: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 3	0 % to 100 % default 50 %	Band 3: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 4	0 % to 100 % default 50 %	Band 4: width of the peak (low = wide, sources share; high = narrow, one owns the band).

CONTROL	RANGE / DEFAULT	WHAT IT DOES
Q 5	0 % to 100 % default 50 %	Band 5: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 6	0 % to 100 % default 50 %	Band 6: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 7	0 % to 100 % default 50 %	Band 7: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 8	0 % to 100 % default 50 %	Band 8: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 9	0 % to 100 % default 50 %	Band 9: width of the peak (low = wide, sources share; high = narrow, one owns the band).
Q 10	0 % to 100 % default 50 %	Band 10: width of the peak (low = wide, sources share; high = narrow, one owns the band).

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

SF TAP

AUDIO EFFECT LATENCY 512 SAMPLES AT 48 KHZ

Puts any track (stems, hardware, other plugins) into Pocket and ALLOC. Adds no sound of its own.



USING IT

— One on each stem track from Live's stem separation, with ROLE set (stem drums = PERC, stem bass = BASS ...).

PERFORM

CONTROL	RANGE / DEFAULT	WHAT IT DOES
ROLE	KICK, BASS, VOICE, LEAD, PERC, RUMBLE ... (9) default OTHER	Priority in ALLOC (KICK highest ... OTHER lowest) and the name shown.
ALLOC	ON / OFF default ON	Take part in ALLOC.
POCKET DEPTH	0 % to 100 % default 0 %	Duck this track around SF KICK.
POCKET SHAPE	0 % to 100 % default 30 %	Pocket shape.
LEVEL	-36.0 dB to 6.0 dB default 0.0 dB	Output level. Default is set so the module sits sensibly in a mix.

M = follows the plugin's MACRO (MACRO SRC / MACRO AMT).

11 APPENDIX

MIDI NOTE MAP

PLUGIN	NOTES	DOES
SF KICK, PERC	any	Plays the sound (KICK: KEY TRACK on makes C1 = TUNE).
SF VOICE	C3, C#3, D3 ...	Slice 1, 2, 3 ... of the phrase. Below C3: the whole phrase.
SF STRIKE	any	Strikes the object (KEY TRACK on: the note sets the pitch, C3 = TUNE).
SF DECK	C1-D#1	Deck A hot cues 1-4 (jump on the next beat).
SF DECK	C2-D#2	Deck B hot cues 1-4.
SF CONTROL	Program Change ch 16	Song 01-16 (channel set by PC CHANNEL), for the AU: Live sends a VST3 none. The show calls songs from the E16 (NEXT SONG, GO).
SF LEAD / BASS ACID	velocity > ~108	Accent. Overlapping notes slide.

WHERE THINGS ARE SAVED

— Songs, samples, ALLOC choices, MOTION random seeds, DECK tracklist and venue EQs: inside the Live set (plugin state).

— User presets: Music/SF Live Suite/Presets/<PLUGIN>, one file per preset.

— DECK track log: Documents/SF Live Suite/tracklog.txt.

— SF TERROIR: its places are inside the Live set; the sweep file is in Music/SF Live Suite/TERROIR, and every take captured in the set (SF CAPTURE) in Music/SF Live Suite/TERROIR/Captures, one WAV each.

— DECK tracks are referenced by file path, not embedded: keep the files where they are.

WHEN SOMETHING ISN'T RIGHT

SYMPTOM	CHECK
SF SHOW.als: a plugin is missing	The suite isn't installed, or VST3 is off (Settings > Plug-Ins). Install or turn VST3 on, then open the set again.
An instrument doesn't play	MIDI From: each track listens to one ONE port and channel (chapter 7). Shown as missing: your Live names the ONE's ports otherwise; set each by the table. Track on for both ONE ports in Live's MIDI preferences. Never All Ins: every instrument would play every part.
A group or track is silent	Its volume (MIX, LEVELS) or ENERGY knob (PERF, NRG) at the far left; SF ALLOC on LIVE BUS with ENABLE on (it cuts lower-priority sources in its bands: ENABLE off to check); the group's output (every group to LIVE BUS).
Nothing ducks	SF CONTROL: BUS SHARED and KICK REPORTING. The kick must be an SF KICK (or use SF DUCK with SOURCE SIDECHAIN).
Pump early or late	Live buffer size 512 or less. Then trim POCKET OFFSET on SF CONTROL by ear.
The ONE doesn't follow Live's tempo	Start from Live, not the ONE: Live sends clock only while it plays. The ONE's header shows USB (stopped: Shift + Stop until it does). Sync on OXI ONE MKII (Port 1)'s output and on no other port. Hold Shift + Back for 2 s: the ONE's MIDI monitor shows the clock arriving while Live plays. Still nothing: Sync on Port 2's output instead, never both.
Songs don't change	From the E16: turn NEXT SONG (encoder 15 of any PERF page) to the song, then press it (GO); SF CONTROL shows NEXT in amber, PRESS GO, until you do. The song must have been stored (SONG nn EMPTY otherwise). Program Change reaches only an AU of SF CONTROL.
E16 rings don't follow	Remote on for E16 Port 1's output in Live's MIDI preferences; map with Live's MIDI Map.
An E16 press does nothing	A scene file from before Nov 8 is loaded: load the current PERF and MIX. A toggle that flips back: Remote is on for two E16 ports, so the note arrives twice. Port 1 only.
An E16 knob jumps when touched	A shared knob holding an old value: controllers/e16 README, check 2. Or Auto Transmit is on: off in the scene and globally.
ALLOC does nothing	ENABLE on, DEPTH above 0 on the band, the instruments' ALLOC toggles on. In DYNAMIC, the favoured source must be playing in that band.
DECK row says MISSING	The file moved. Drop it again.
A knob shows GLIDE	SLOW ALL or the plugin's SLOW is on: the knob is travelling to where you set it.
A knob moves on its own	MOTION: check M1 / M2 TARGET (the plugin's MOTION group; its title line says where they go).
Header shows FAULT (red)	The plugin caught an invalid value or a runaway and reset itself to silence. It carries on working. If it happens again, note what was playing: it is a bug.
SF TERROIR won't capture	Its status line says why. The transport must be stopped; NO SF CAPTURE: put it on the room mics' track (ROOM in SF SHOW), its device on; one take at a time.
A capture says THE SWEEP DIDN'T COME BACK, or under 30 dB	The speaker didn't play the sweep, or the mics didn't hear it: the speaker on ROOM's output (3/4), the track on and up; the mics on ROOM's input (Input Config: 1/2 a stereo pair), Monitor In. Then louder or closer. The slot stays as it was.
A capture says NO SWEEP IN THE RIGHT CHANNEL	That mic heard nothing (its cable, its gain, Input Config): the place came in mono from the other. Fix it and capture again.
A capture says CAPTURE STOPPED: SF CAPTURE STOPPED ANSWERING	ROOM's SF CAPTURE was removed or switched off, or its track's audio stood still. Put it back and capture again: nothing was stored.
Something won't stop	The audio interface's output knob down: the room goes quiet, the recording carries on. Then find it. PANIC on SF CONTROL, held in its window, fades every SF plugin out and clears every voice, tail and delay line; Live's own clips and devices aren't SF.

SYMPTOM	CHECK
Header shows PANIC (red)	PANIC is on at SF CONTROL: held in its window, or switched on by a mapping. Let it go there.
CONTROL shows LOCAL ONLY	The plugins come from different builds. Rebuild and install the whole suite.
A return doubles the dry signal	DRY off on SF DELAY, SF REVERB or SF DENSITY on a return track.
Pitch is wrong after a song change	KEY on SF CONTROL is stored per song. KEY FOLLOW off on an instrument ignores it.
Levels changed after the update	Since Nov 3 every model is calibrated to its role's target (chapter 5). Re-balance the faders once.

E16 ADDRESS MAP (PERF, MIX)

Every address of the two E16 scenes once, by channel: the Live target, the mapping's range, and where the knob is on the E16 (a PERF page is named by its track's code; LEVELS, SPACE and MACROS are MIX's). SF SHOW.als has them all mapped; this list is for checking it, and for mapping a set of your own (Cmd-M). The same list is controllers/e16/sf-show-map.csv.

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
1	CC 12	1 KICK > SF KICK > TUNE	absolute	KCK 1
1	CC 13	1 KICK > SF KICK > PUNCH	absolute	KCK 2
1	CC 14	1 KICK > SF KICK > CLICK	absolute	KCK 3
1	CC 15	1 KICK > SF KICK > DRIVE	absolute	KCK 4
1	CC 16	1 KICK > SF KICK > PITCH TIME	absolute	KCK 5
1	CC 17	1 KICK > SF KICK > LAYER	absolute	KCK 6
1	CC 18	1 KICK > SF KICK > ECHO SPACE	absolute	KCK 7
1	CC 19	1 KICK > SF KICK > ECHO SWING	absolute	KCK 8
1	CC 20	1 KICK > SF KICK > DECAY	absolute	KCK 9
1	CC 21	1 KICK > SF KICK > DENSITY	absolute	KCK 10
1	CC 22	1 KICK > SF KICK > FILTER	absolute	KCK 11
1	CC 23	1 KICK > SF KICK > LEVEL	absolute	KCK 12
1	CC 28	1 KICK > SF ENERGY > ENERGY	absolute	NRG 1
1	CC 102	1 KICK > mixer > volume	absolute, Max 0 dB	LEVELS 1
1	CC 103	1 KICK > mixer > send A	absolute, Max 0 dB	KCK 13
1	CC 104	1 KICK > mixer > send B	absolute, Max 0 dB	KCK 14
2	CC 12	2 CLOSED HAT > SF PERC > TUNE	absolute	CHH 1
2	CC 13	2 CLOSED HAT > SF PERC > TONE	absolute	CHH 2
2	CC 14	2 CLOSED HAT > SF PERC > ACCENT	absolute	CHH 3
2	CC 15	2 CLOSED HAT > SF PERC > DRIVE	absolute	CHH 4
2	CC 16	2 CLOSED HAT > SF PERC > ECHO SPACE	absolute	CHH 5
2	CC 17	2 CLOSED HAT > SF PERC > ECHO SWING	absolute	CHH 6
2	CC 18	2 CLOSED HAT > SF PERC > ECHO PROB	absolute	CHH 7
2	CC 19	2 CLOSED HAT > SF PERC > SPREAD	absolute	CHH 8
2	CC 20	2 CLOSED HAT > SF PERC > DECAY	absolute	CHH 9
2	CC 21	2 CLOSED HAT > SF PERC > DENSITY	absolute	CHH 10
2	CC 22	2 CLOSED HAT > SF PERC > FILTER	absolute	CHH 11

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
2	CC 23	2 CLOSED HAT > SF PERC > LEVEL	absolute	CHH 12
2	CC 28	2 CLOSED HAT > SF ENERGY > ENERGY	absolute	NRG 2
2	CC 102	2 CLOSED HAT > mixer > volume	absolute, Max 0 dB	LEVELS 2
2	CC 103	2 CLOSED HAT > mixer > send A	absolute, Max 0 dB	CHH 13
2	CC 104	2 CLOSED HAT > mixer > send B	absolute, Max 0 dB	CHH 14
3	CC 12	3 OPEN HAT > SF PERC > TUNE	absolute	OHH 1
3	CC 13	3 OPEN HAT > SF PERC > TONE	absolute	OHH 2
3	CC 14	3 OPEN HAT > SF PERC > ACCENT	absolute	OHH 3
3	CC 15	3 OPEN HAT > SF PERC > DRIVE	absolute	OHH 4
3	CC 16	3 OPEN HAT > SF PERC > ECHO SPACE	absolute	OHH 5
3	CC 17	3 OPEN HAT > SF PERC > ECHO SWING	absolute	OHH 6
3	CC 18	3 OPEN HAT > SF PERC > ECHO PROB	absolute	OHH 7
3	CC 19	3 OPEN HAT > SF PERC > SPREAD	absolute	OHH 8
3	CC 20	3 OPEN HAT > SF PERC > DECAY	absolute	OHH 9
3	CC 21	3 OPEN HAT > SF PERC > DENSITY	absolute	OHH 10
3	CC 22	3 OPEN HAT > SF PERC > FILTER	absolute	OHH 11
3	CC 23	3 OPEN HAT > SF PERC > LEVEL	absolute	OHH 12
3	CC 28	3 OPEN HAT > SF ENERGY > ENERGY	absolute	NRG 3
3	CC 102	3 OPEN HAT > mixer > volume	absolute, Max 0 dB	LEVELS 3
3	CC 103	3 OPEN HAT > mixer > send A	absolute, Max 0 dB	OHH 13
3	CC 104	3 OPEN HAT > mixer > send B	absolute, Max 0 dB	OHH 14
4	CC 12	4 CLAP > SF PERC > TUNE	absolute	CLP 1
4	CC 13	4 CLAP > SF PERC > TONE	absolute	CLP 2
4	CC 14	4 CLAP > SF PERC > ACCENT	absolute	CLP 3
4	CC 15	4 CLAP > SF PERC > DRIVE	absolute	CLP 4
4	CC 16	4 CLAP > SF PERC > ECHO SPACE	absolute	CLP 5
4	CC 17	4 CLAP > SF PERC > ECHO SWING	absolute	CLP 6
4	CC 18	4 CLAP > SF PERC > ECHO PROB	absolute	CLP 7
4	CC 19	4 CLAP > SF PERC > SPREAD	absolute	CLP 8
4	CC 20	4 CLAP > SF PERC > DECAY	absolute	CLP 9
4	CC 21	4 CLAP > SF PERC > DENSITY	absolute	CLP 10
4	CC 22	4 CLAP > SF PERC > FILTER	absolute	CLP 11
4	CC 23	4 CLAP > SF PERC > LEVEL	absolute	CLP 12
4	CC 28	4 CLAP > SF ENERGY > ENERGY	absolute	NRG 4
4	CC 102	4 CLAP > mixer > volume	absolute, Max 0 dB	LEVELS 4
4	CC 103	4 CLAP > mixer > send A	absolute, Max 0 dB	CLP 13
4	CC 104	4 CLAP > mixer > send B	absolute, Max 0 dB	CLP 14
5	CC 12	5 PERC > SF STRIKE > POWER	absolute	PRC 1
5	CC 13	5 PERC > SF STRIKE > DISTANCE	absolute	PRC 2
5	CC 14	5 PERC > SF STRIKE > SIZE	absolute	PRC 3
5	CC 15	5 PERC > SF STRIKE > MATERIAL	absolute	PRC 4
5	CC 16	5 PERC > SF STRIKE > POSITION	absolute	PRC 5
5	CC 17	5 PERC > SF STRIKE > TENSION	absolute	PRC 6

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
5	CC 18	5 PERC > SF STRIKE > SPREAD	absolute	PRC 7
5	CC 19	5 PERC > SF STRIKE > TUNE	absolute	PRC 8
5	CC 20	5 PERC > SF STRIKE > DECAY	absolute	PRC 9
5	CC 21	5 PERC > SF STRIKE > DENSITY	absolute	PRC 10
5	CC 22	5 PERC > SF STRIKE > FILTER	absolute	PRC 11
5	CC 23	5 PERC > SF STRIKE > LEVEL	absolute	PRC 12
5	CC 28	5 PERC > SF ENERGY > ENERGY	absolute	NRG 5
5	CC 102	5 PERC > mixer > volume	absolute, Max 0 dB	LEVELS 5
5	CC 103	5 PERC > mixer > send A	absolute, Max 0 dB	PRC 13
5	CC 104	5 PERC > mixer > send B	absolute, Max 0 dB	PRC 14
6	CC 12	6 BASS > SF BASS > CUTOFF	absolute	BAS 1
6	CC 13	6 BASS > SF BASS > RES	absolute	BAS 2
6	CC 14	6 BASS > SF BASS > ENV AMT	absolute	BAS 3
6	CC 15	6 BASS > SF BASS > DECAY	absolute	BAS 4
6	CC 16	6 BASS > SF BASS > DRIVE	absolute	BAS 5
6	CC 17	6 BASS > SF BASS > ACCENT	absolute	BAS 6
6	CC 18	6 BASS > SF BASS > GLIDE	absolute	BAS 7
6	CC 19	6 BASS > SF BASS > POCKET DEPTH	absolute	BAS 8
6	CC 20	6 BASS > SF BASS > GATE	absolute	BAS 9
6	CC 21	6 BASS > SF BASS > DENSITY	absolute	BAS 10
6	CC 22	6 BASS > SF BASS > FILTER	absolute	BAS 11
6	CC 23	6 BASS > SF BASS > LEVEL	absolute	BAS 12
6	CC 28	6 BASS > SF ENERGY > ENERGY	absolute	NRG 6
6	CC 102	6 BASS > mixer > volume	absolute, Max 0 dB	LEVELS 6
6	CC 103	6 BASS > mixer > send A	absolute, Max 0 dB	BAS 13
6	CC 104	6 BASS > mixer > send B	absolute, Max 0 dB	BAS 14
7	CC 12	7 RUMBLE > SF RUMBLE > SIZE	absolute	RMB 1
7	CC 13	7 RUMBLE > SF RUMBLE > DARK	absolute	RMB 2
7	CC 14	7 RUMBLE > SF RUMBLE > GRIT	absolute	RMB 3
7	CC 15	7 RUMBLE > SF RUMBLE > PUMP	absolute	RMB 4
7	CC 16	7 RUMBLE > SF RUMBLE > TUNE	absolute	RMB 5
7	CC 17	7 RUMBLE > SF RUMBLE > PREDELAY	absolute	RMB 6
7	CC 18	7 RUMBLE > SF RUMBLE > WIDTH	absolute	RMB 7
7	CC 19	7 RUMBLE > SF RUMBLE > POCKET SHAPE	absolute	RMB 8
7	CC 20	7 RUMBLE > SF RUMBLE > GATE	absolute	RMB 9
7	CC 21	7 RUMBLE > SF RUMBLE > DENSITY	absolute	RMB 10
7	CC 22	7 RUMBLE > SF RUMBLE > FILTER	absolute	RMB 11
7	CC 23	7 RUMBLE > SF RUMBLE > LEVEL	absolute	RMB 12
7	CC 28	7 RUMBLE > SF ENERGY > ENERGY	absolute	NRG 7
7	CC 102	7 RUMBLE > mixer > volume	absolute, Max 0 dB	LEVELS 7
7	CC 103	7 RUMBLE > mixer > send A	absolute, Max 0 dB	RMB 13
7	CC 104	7 RUMBLE > mixer > send B	absolute, Max 0 dB	RMB 14
8	CC 12	8 LEAD > SF LEAD > CUTOFF	absolute	LED 1

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
8	CC 13	8 LEAD > SF LEAD > RES	absolute	LED 2
8	CC 14	8 LEAD > SF LEAD > ENV AMT	absolute	LED 3
8	CC 15	8 LEAD > SF LEAD > DECAY	absolute	LED 4
8	CC 16	8 LEAD > SF LEAD > DRIVE	absolute	LED 5
8	CC 17	8 LEAD > SF LEAD > ACCENT	absolute	LED 6
8	CC 18	8 LEAD > SF LEAD > GLIDE	absolute	LED 7
8	CC 19	8 LEAD > SF LEAD > COLOR	absolute	LED 8
8	CC 20	8 LEAD > SF LEAD > GATE	absolute	LED 9
8	CC 21	8 LEAD > SF LEAD > DENSITY	absolute	LED 10
8	CC 22	8 LEAD > SF LEAD > FILTER	absolute	LED 11
8	CC 23	8 LEAD > SF LEAD > LEVEL	absolute	LED 12
8	CC 28	8 LEAD > SF ENERGY > ENERGY	absolute	NRG 8
8	CC 29	8 LEAD > SF LEAD > ACID	absolute	NRG 12
8	CC 102	8 LEAD > mixer > volume	absolute, Max 0 dB	LEVELS 8
8	CC 103	8 LEAD > mixer > send A	absolute, Max 0 dB	LED 13
8	CC 104	8 LEAD > mixer > send B	absolute, Max 0 dB	LED 14
9	CC 12	9 CHORD > SF LEAD > CUTOFF	absolute	CRD 1
9	CC 13	9 CHORD > SF LEAD > RES	absolute	CRD 2
9	CC 14	9 CHORD > SF LEAD > ENV AMT	absolute	CRD 3
9	CC 15	9 CHORD > SF LEAD > DECAY	absolute	CRD 4
9	CC 16	9 CHORD > SF LEAD > ATTACK	absolute	CRD 5
9	CC 17	9 CHORD > SF LEAD > SPACE	absolute	CRD 6
9	CC 18	9 CHORD > SF LEAD > DETUNE	absolute	CRD 7
9	CC 19	9 CHORD > SF LEAD > POCKET DEPTH	absolute	CRD 8
9	CC 20	9 CHORD > SF LEAD > GATE	absolute	CRD 9
9	CC 21	9 CHORD > SF LEAD > DENSITY	absolute	CRD 10
9	CC 22	9 CHORD > SF LEAD > FILTER	absolute	CRD 11
9	CC 23	9 CHORD > SF LEAD > LEVEL	absolute	CRD 12
9	CC 28	9 CHORD > SF ENERGY > ENERGY	absolute	NRG 9
9	CC 102	9 CHORD > mixer > volume	absolute, Max 0 dB	LEVELS 9
9	CC 103	9 CHORD > mixer > send A	absolute, Max 0 dB	CRD 13
9	CC 104	9 CHORD > mixer > send B	absolute, Max 0 dB	CRD 14
10	CC 12	10 PAD > SF PAD > CUTOFF	absolute	PAD 1
10	CC 13	10 PAD > SF PAD > RES	absolute	PAD 2
10	CC 14	10 PAD > SF PAD > COLOR	absolute	PAD 3
10	CC 15	10 PAD > SF PAD > DETUNE	absolute	PAD 4
10	CC 16	10 PAD > SF PAD > ATTACK	absolute	PAD 5
10	CC 17	10 PAD > SF PAD > DECAY	absolute	PAD 6
10	CC 18	10 PAD > SF PAD > SPACE	absolute	PAD 7
10	CC 19	10 PAD > SF PAD > POCKET DEPTH	absolute	PAD 8
10	CC 20	10 PAD > SF PAD > GATE	absolute	PAD 9
10	CC 21	10 PAD > SF PAD > DENSITY	absolute	PAD 10
10	CC 22	10 PAD > SF PAD > FILTER	absolute	PAD 11

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
10	CC 23	10 PAD > SF PAD > LEVEL	absolute	PAD 12
10	CC 28	10 PAD > SF ENERGY > ENERGY	absolute	NRG 10
10	CC 29	10 PAD > SF PAD > WARP	absolute	NRG 14
10	CC 102	10 PAD > mixer > volume	absolute, Max 0 dB	LEVELS 10
10	CC 103	10 PAD > mixer > send A	absolute, Max 0 dB	PAD 13
10	CC 104	10 PAD > mixer > send B	absolute, Max 0 dB	PAD 14
11	CC 12	11 VOICE > SF VOICE > SIGNAL	absolute	VOX 1
11	CC 13	11 VOICE > SF VOICE > TONE	absolute	VOX 2
11	CC 14	11 VOICE > SF VOICE > TUNE	absolute	VOX 3
11	CC 15	11 VOICE > SF VOICE > DRIVE	absolute	VOX 4
11	CC 16	11 VOICE > SF VOICE > ECHO	absolute	VOX 5
11	CC 17	11 VOICE > SF VOICE > ECHO TIME	absolute	VOX 6
11	CC 18	11 VOICE > SF VOICE > ECHO FDBK	absolute	VOX 7
11	CC 19	11 VOICE > SF VOICE > POCKET DEPTH	absolute	VOX 8
11	CC 20	11 VOICE > SF VOICE > GATE	absolute	VOX 9
11	CC 21	11 VOICE > SF VOICE > DENSITY	absolute	VOX 10
11	CC 22	11 VOICE > SF VOICE > FILTER	absolute	VOX 11
11	CC 23	11 VOICE > SF VOICE > LEVEL	absolute	VOX 12
11	CC 28	11 VOICE > SF ENERGY > ENERGY	absolute	NRG 11
11	CC 102	11 VOICE > mixer > volume	absolute, Max 0 dB	LEVELS 11
11	CC 103	11 VOICE > mixer > send A	absolute, Max 0 dB	VOX 13
11	CC 104	11 VOICE > mixer > send B	absolute, Max 0 dB	VOX 14
12	CC 44	A DUB > SF DELAY > TIME	absolute	SPACE 1
12	CC 45	A DUB > SF DELAY > FEEDBACK	absolute, 0-70 % at first	SPACE 2
12	CC 46	A DUB > SF DELAY > TONE	absolute	SPACE 3
12	CC 47	A DUB > SF DELAY > DRIVE	absolute	SPACE 4
12	CC 60	A DUB > SF DELAY > THROW	press: hold	SPACE 1
12	CC 102	A DUB > mixer > volume	absolute, Max 0 dB	LEVELS 15
12	NOTE 61 (C#3)	A DUB > SF DELAY > FREEZE	press: toggle	SPACE 2
13	CC 48	B SPACE > SF REVERB > SIZE	absolute	SPACE 5
13	CC 49	B SPACE > SF REVERB > DECAY	absolute	SPACE 6
13	CC 50	B SPACE > SF REVERB > DARK	absolute	SPACE 7
13	CC 51	B SPACE > SF REVERB > DUCK	absolute	SPACE 8
13	CC 64	B SPACE > SF REVERB > THROW	press: hold	SPACE 5
13	CC 102	B SPACE > mixer > volume	absolute, Max 0 dB	LEVELS 16
13	NOTE 65 (F3)	B SPACE > SF REVERB > FREEZE	press: toggle	SPACE 6
14	CC 52	DRUMS > SF REPEAT > LENGTH	absolute	MACROS 9
14	CC 53	DRUMS > SF REPEAT > PITCH	absolute	MACROS 10
14	CC 68	DRUMS > SF REPEAT > ROLL	press: hold	MACROS 9
14	CC 102	DRUMS > mixer > volume	absolute, Max 0 dB	LEVELS 12
14	CC 103	LOW > mixer > volume	absolute, Max 0 dB	LEVELS 13
14	CC 104	MUSIC > mixer > volume	absolute, Max 0 dB	LEVELS 14

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
15	CC 27	LIVE BUS > SF BUILD > BUILD	absolute	every PERF page, encoder 16
15	CC 29	LIVE BUS > SF DRIVE > MIX	absolute	NRG 13
15	CC 43	LIVE BUS > SF BUILD > DROP	press: hold	every PERF page, encoder 16
16	CC 12	CONTROL > SF CONTROL > SLOW ALL	absolute	every PERF page, encoder 1 (second destination), MACROS 7
16	CC 26	CONTROL > SF CONTROL > NEXT SONG	absolute	every PERF page, encoder 15
16	CC 42	CONTROL > SF CONTROL > GO	press: hold	every PERF page, encoder 15
16	CC 44	CONTROL > SF CONTROL > MACRO 1	absolute	MACROS 1
16	CC 45	CONTROL > SF CONTROL > MACRO 2	absolute	MACROS 2
16	CC 46	CONTROL > SF CONTROL > MACRO 3	absolute	MACROS 3
16	CC 47	CONTROL > SF CONTROL > MACRO 4	absolute	MACROS 4
16	CC 48	CONTROL > SF CONTROL > MACRO 5	absolute	MACROS 5
16	CC 49	CONTROL > SF CONTROL > MACRO 6	absolute	MACROS 6